1 Overview of Data Collection and Information Functions for Quality Data on the AIP

Purpose

The CAQ basic functionality provides the inspection steps to perform inspections. In the combined BDE + CAQ operation mode, the system identifies the respective inspection step when you log on an operation or the system identifies the inspection plan for an article and operation number. Using the inspection plan, the inspection step is created and logged on with the operation. Different events can trigger an inspection (time and piece interval, machine status change, etc.).

Depending on its configuration, you can also use the AIP as an exclusive inspection station to record inspection data.

In addition to the inspections in production, you also use the AIP to perform inspections in the goods receipt and goods issue, to inspect the initial sample and to perform calibrations.

1.1 Integration in the main view

If a terminal is operated as a QM terminal, the main view of the AIP2 terminal also shows CAQ data. The machine and order list show the **inspection status** graphically or in text form:

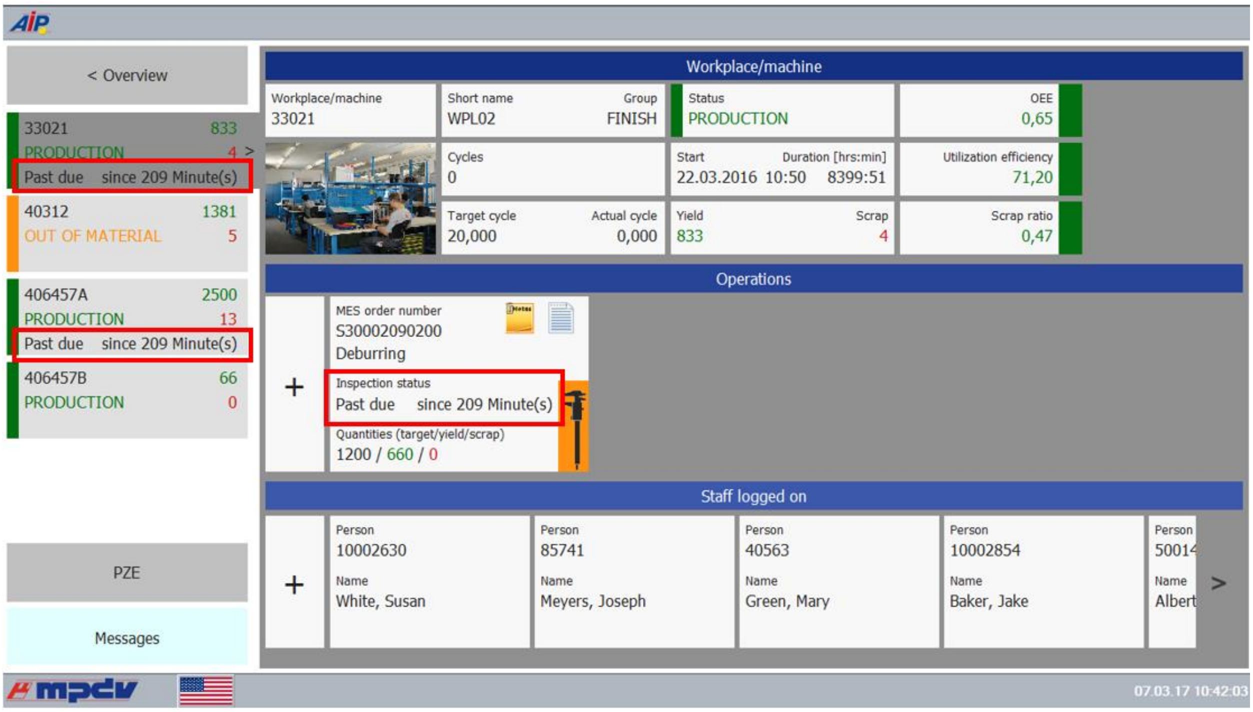


Figure: Main view if QM operation is enabled

If the inspection point list is used that is provided with FEP, WEP, PMV 8.2, the inspection statuses are not shown. With this configuration, the inspection points are only requested when the inspection point list is requested.

1.1.1 Due date status

Only the main view for machines and orders shows the due date status.

The due date status specifies if inspection specifications have been reached. The due date status does not depend on user rights or any other user-dependent limitations.

The due date status is made up of three information components:

- Colored inspection status: due date status as a symbol
- Inspection status in text form: due date status in plain text
- Time

There is a separate due date status used to display incorrect configurations:

	Incorrect configuration (e.g. with unknown evaluation basis or inspection sequence)	**Error**

The following colors are available to display other due date statuses:

	Entry required to reach the minimum inspections	**due for x minute(s)**
	Minimum inspection scope reached	**checked**
	Minimum inspection scope not reached	**checked**
	Inspection step or inspection requirement completed	**completed**

With attributive characteristics, the minimum inspection scope is reached if the number of inspected parts matches the inspection scope. This is based on the default configuration "less than" for the inspection scope indicator. The inspection scope indicator is defined in the CAQ system option 1176.

If the sample size is not defined (0 or empty), the minimum inspection scope is never reached. In this case, the color of the node point is light green.

With cavity characteristics, the inspection scope indicator is always set to "EGAL". The color of the node point therefore does not change and is light green.

1.1.2 Special features in case of shift change and queue mode

During a change of shifts, most of the requests sent to the HYDRA server are collected in a queue and processed in the same order after the shift change. The same processing applies for CAQ inspection activities such as saving a measured value, creating an inspection point, etc. If these activities have an effect on the structure of the inspection list (generation and completion of inspections points), this will not affect the shift change.

And for example, inspection points generated on the server during shift change cannot be seen on the AIP, because in this case communication with the server is limited.

The system uses two kinds of messages to indicate if a queue is available (recognizable by the underlined number in the footer next to the AIP status) or if all stored commands in the queue have been processed.

1.1.2.1 Commands in queue mode

For some activities that are run in queue mode, a message is displayed that informs about the available restrictions. By default, the following messages are output:

Generation of inspection points, completion of inspection points

Communication with the HYDRA server is currently limited. Data will be updated when server communication is running again and when no more dialogs are open.

Show info

Communication with the HYDRA server is currently limited. Data might not be current.

Update display

Communication with the HYDRA server is currently limited. Data cannot be updated.

☞ **Configuration**

Buttons of the inspection list

Configure individual settings for message texts or enable and disable messages in the configuration file `caq_dc_t.ini`.

1.1.2.2 Action when exiting queue mode

CAQ data is automatically updated when the queue mode is closed.

☞ **Configuration**

CAQ action after queue mode

Configure individual settings for message texts or enable and disable messages in the configuration file `caq72.ini.`

1.2 Inspector ID

Depending on the terminal configuration, the terminal can show a dialog to identify the inspecting person before the dialog to record inspection results is opened.

You can therefore record inspection results in two different user modes. The system either uses so-called *constant* or *changing* users.

1.2.1 Constant user

Before opening the inspection results recording, the system asks for the staff badge number to identify the inspecting person.

The staff badge number entered here is constant for all input dialogs that follow and can no longer be changed in the input dialogs of inspection results.

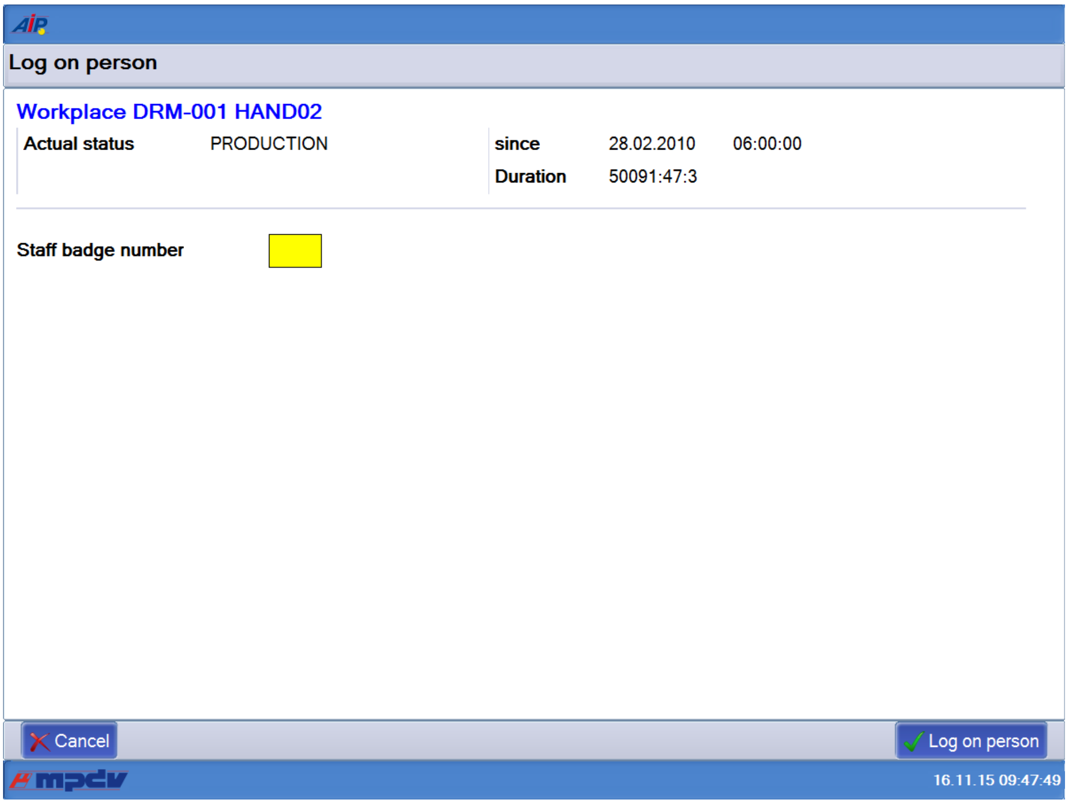


Figure: Dialog to enter the staff badge number

Configuration

Configure the dialog to enter the staff badge number on the MOC. Go to: *File* → *Status information* → *Terminal status*. To do this, enable the option *Inspector identification before opening inspection dialog* in the *CAQ* tab.

You can extend this query function if you also enable the sub-option "*Only if inspector is unknown*". This setting is only useful with a *constant user*.

1.2.2 Changing user

The system does not ask for the staff badge number when you open the recording of inspection results. Enter the staff badge number manually in each dialog if needed.
In some cases, a validation check of the user or of the user's rights might be run directly on the server or in a dialog.

1.3 Recording of inspection results

You use the recording of inspection results to collect QM data on the terminal. The inspection results recording also provides features allowing to present graphics (e.g. control charts) and to prepare and present additional information, such as statistical KPIs, for example.

The dialog of the inspection results recording is divided into two windows.

The left hand side permanently shows the so-called *inspection list* with the respective action elements. The inspection list is presented in a tree structure. The tree structure is always entirely expanded. The structure cannot be collapsed.

The right hand side shows the *input panel* with the respective action elements. This panel shows the detail data for the element selected in the inspection list. The data is with or without write protection (read-only mode), depending on the mode configured.

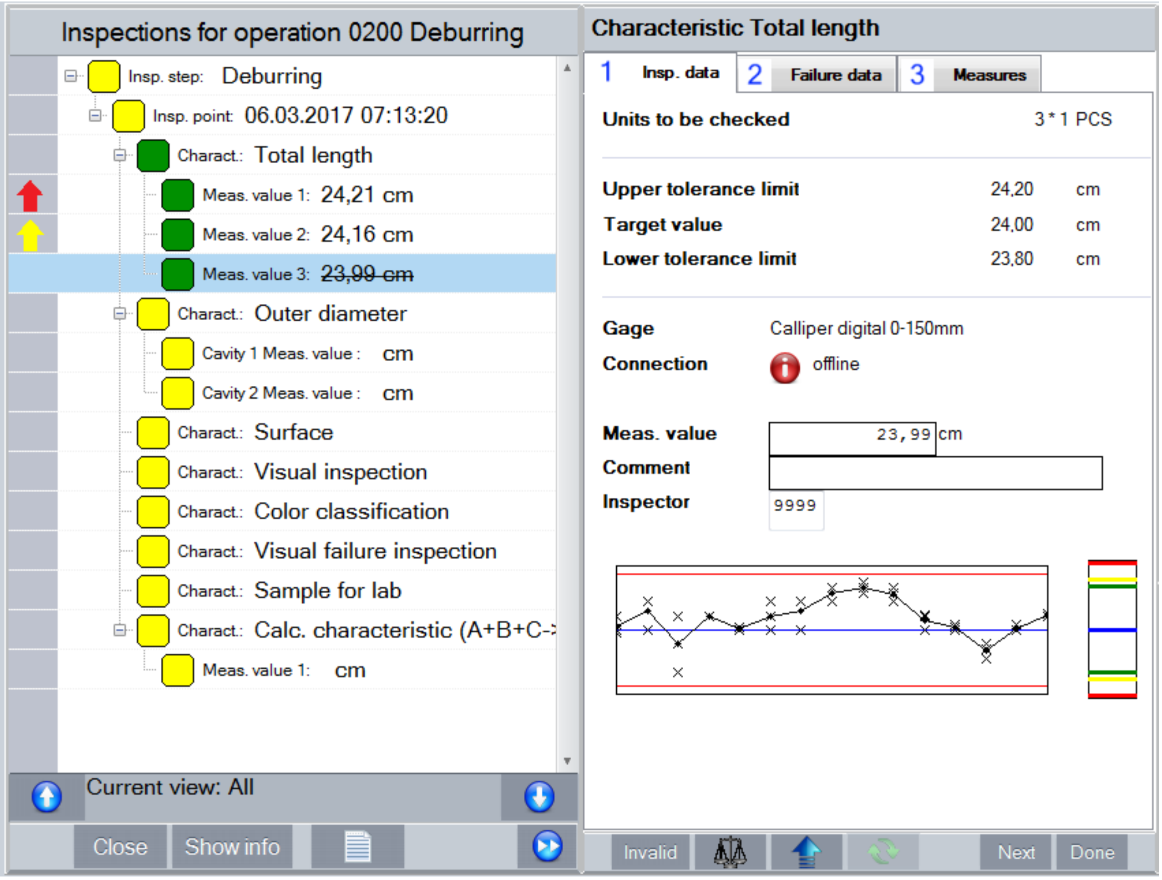


Figure: Recording of inspection results - the left hand side always shows the *inspection list*, the right hand side shows a variable *input panel*

☞
Note

Recording of inspection results with many inspection points

In the recording of inspection results, there are usually few inspection points for the operation logged on. An increasing number of inspection points and a great number of inspection point characteristics have a negative impact on the performance.

For this purpose, the system provides the function of the "preceding inspection point list" as of FEP, WEP, PMV 8.2. The preceding inspection point list is opened before the actual inspection list is called. You use this list to select the inspection point that you want to check. Consequently, the contents of the inspection list are reduced to one inspection point.

1.3.1 The inspection list

The inspection list consists of individual action elements (= tree nodes).

☞
Note

Buttons of the inspection list

If you switch back and forth between different action elements, you change the input area and the buttons for the inspection list.

1.3.1.1 Buttons of the inspection list

The buttons are located at the bottom of the inspection list. Depending on the active action element, the displayed buttons offer different functions matching the relevant context.

☞
Configuration

Buttons of the inspection list

Configure the buttons individually in the configuration file `caq_dc_t.ini`.

1.3.1.2 *Page 1 - context-sensitive buttons*

The buttons on *page 1* contain the functions used most frequently.

They depend exclusively on the type of action element currently selected in the inspection list.

One button on *page 1* does not depend on the context. You can use this button to exit the recording of inspection results.

"Close" button

You can use this button to exit the inspection results recording and go back to the machine overview.

Figure: Page 1 - context-independent button "Close"

"New inspection point" button

If you select an inspection point or an inspection step, which is relevant to an inspection point, in the inspection list, the inspection list also shows the button *New inspection point*:

Figure: Page 1 - context-dependent button "New inspection point"

Click this button to manually create a new inspection point. The newly created inspection point then appears in the inspection list.

"New measurement" button

If you select a characteristic or a single value (individual piece inspection) in the inspection list, the button *New measurement* is shown at the bottom of the inspection list:

Figure: Page 1 - context-dependent button "New measurement"

Click this button to manually create an action element for a new measurement.

"Show info" button

If you select a characteristic or a single value (active individual piece inspection) in the inspection list, the button *Show info* is also shown at the bottom of the inspection list:

Figure: Page 1 - context-dependent button "Show info"

By clicking on this button, a screen appears with information on the characteristic.

1.3.1.3 *Page 2 - other functions [update]*

The system provides one or more buttons in order to update/refresh data. The relevant function specifies which data is updated when clicking this button.

Figure: Page 2 - button "Update display"

The function *"update display"* provides three different processing options for the provision of data on the terminal. The following sections describe these options:

1) Function "update all"

The system updates the data of all inspection steps available on the terminal.

This function is useful, for example, if you want to view the recently recorded inspection results of other workplaces.

This function can be useful if a user was absent over a longer period of time and the user wants to update all QM data. Using this function, the user need not repeat the update in every inspection list.

Configuration	**Terminal configuration file `caq_dc_t.ini / caq_dc_t.*`**
	• Layout definitions for buttons of the CAQ inspection list, e.g. here in relation to the inspection point
	`[CAQ_DC_T-PPKT-Page2]`
	`1=DQC_RELOAD_LEGACY,R,Anzeige aktualisieren`
	➔ Default caq_dc_t.dll version <= 2.0.2.50 (AIP 8.1)

2) Function "update inspection point"

The system refreshes the data of the currently selected inspection point.

Use this function to request calculated inspection results from the server for characteristics calculated externally.

Configuration

Terminal configuration file `caq_dc_t.ini / caq_dc_t.*`

- Layout definitions for buttons of the CAQ inspection list, e.g. here in relation to the inspection point

```
[CAQ_DC_T-PPKT-Page2]
1=DQC_RELOAD_PPKT,R,Anzeige aktualisieren
```

➔ Default caq_dc_t.dll version >=2.0.2.51 (AIP 8.1)
➔ Default caq_dc_t.dll version >=8.0.2.7 (AIP 8.2)

3) Function "intelligent update"

This function identifies in which mode[1] you opened the inspection list and the context of the selected node.

According to the criteria mentioned above, the system selects one of the above-described functions 1) or 2).

- context = inspection point or
  context = any node below an inspection point
  ➔ executes the **function "update inspection point"**

- context <> inspection point
  ➔ executes **the function "update all"**

Configuration

Terminal configuration file `caq_dc_t.ini / caq_dc_t.*`

- Layout definitions for buttons of the CAQ inspection list, e.g. here in relation to the inspection point

```
[CAQ_DC_T-PPKT-Page2]
1=DQC_RELOAD,R, update display
```

➔ Default caq_dc_t.dll version >=2.0.2.51 (AIP 8.1)
➔ Default caq_dc_t.dll version >=8.0.2.7 (AIP 8.2)

[1] mode a) standard

mode b) preceding inspection list

You can also integrate several buttons with update functions. The user can then update different data.

The default configuration always provides one button to update data. This button does not depend on the node currently selected in the inspection list. The button then executes the function "intelligent update".

☞ **Note**	**Recording of inspection results with many inspection points** If you operate the terminal in the mode `QUEUE_MODE_RELOAD`, you cannot update data as described above.

When you update the display, the system requests all inspection data once more from the HYDRA server. Depending on the data volume, the action of requesting lists, filling the data structure and structuring the inspection list can take quite a bit of time.

☞ **Note**	**"Update display" button** You cannot specify a concrete time that the update requires. The data volume that must be updated has a big influence. Requirement: The hardware and software conditions must be fulfilled and the HYDRA server's response time must be within the normal range.

1.3.1.4 Collection status

You can identify the respective collection status via the color of the symbol that is placed before the action element in the inspection list.

The following options are available:

There is a separate collection status used to display incorrect configurations:

- ⛔ Incorrect configuration
  (e.g. with unknown evaluation basis or inspection sequence)

The following entries are available to show other collection statuses:

- 🟨 Data collection required
- 🟩 Further data can be recorded

-  No further data can be recorded but corrected
-  Completed

1.3.2 The input panel

The input panel is different for the different action elements. Depending on the selected action element, the application loads and displays the correct dialog including data.

Some of the dialogs are only for information purposes, others can be used to record data. If the dialog is an input dialog, the write permission must be available, if required. A dialog can also be read-only, because additional processing is no longer allowed.

Similar to the inspection list, the different dialogs provide different buttons. The buttons provided depend on the functionalities provided by the dialog.

1.4 General input functions

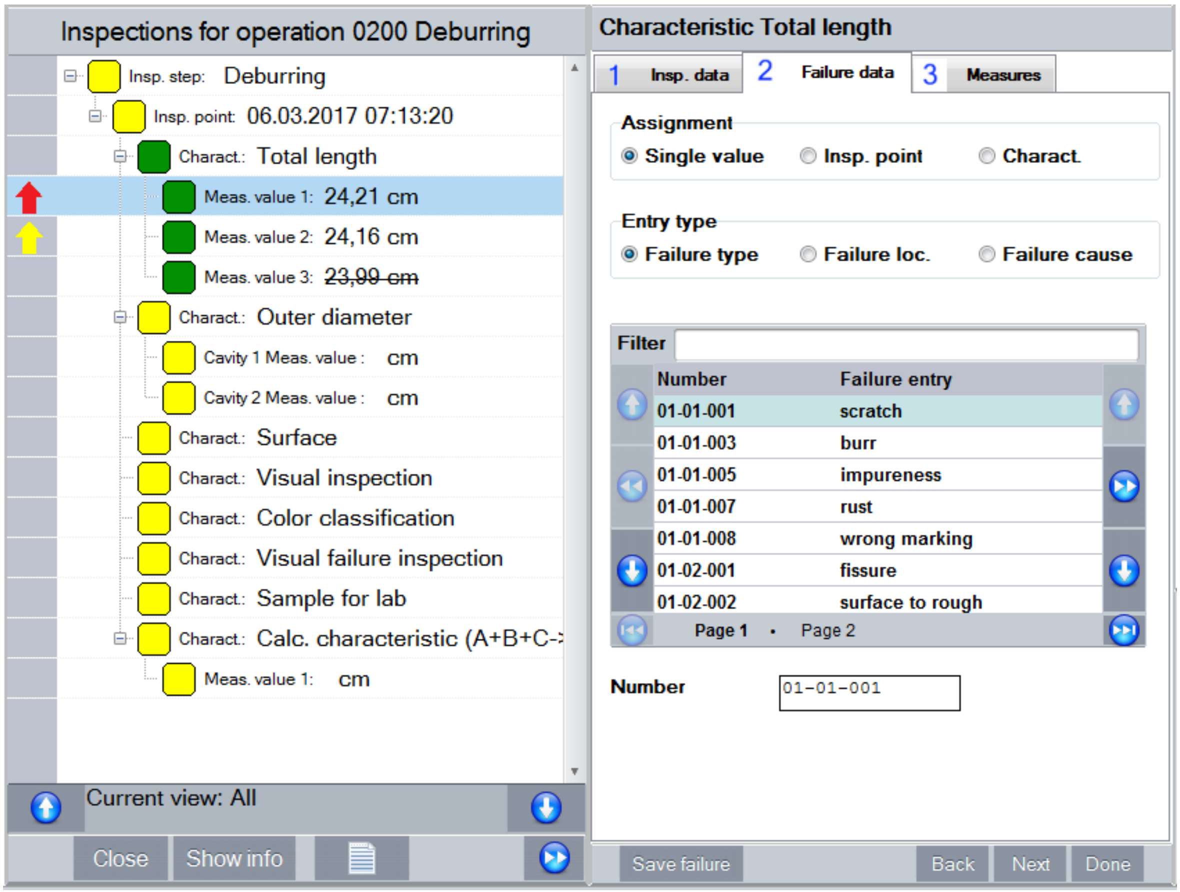
The *input panel* shows the input functions. Presentation and number of available *input functions* are different.

The *input functions* can be included in one or several tabs. At this point, we refer to so-called *workflows*.

The selected action element specifies which input functions are displayed.

1.4.1 Recording of failures

The input dialog *Classic recording of failures* is integrated in many different input functions. Depending on the context, different data is assembled.



☞
Note

Assignment

The items offered for selection in the radio group depend on the context and therefore they are made available dynamically.

☞
Note

Input type

By default, the radio group provides the following items:

- *Failure type*
- *Failure location*
- *Failure cause*

There are recording functions such as the *Inspection chart* where the *input types* must be dynamically assembled.

- If an *analysis selection catalog* is defined for a characteristic, this catalog specifies the possible entries.

- Enter a filter to restrict the failures displayed for selection in the field *Input type*.

Behavior and operation of the *Classic recording of failures* is the same in all dialogs.

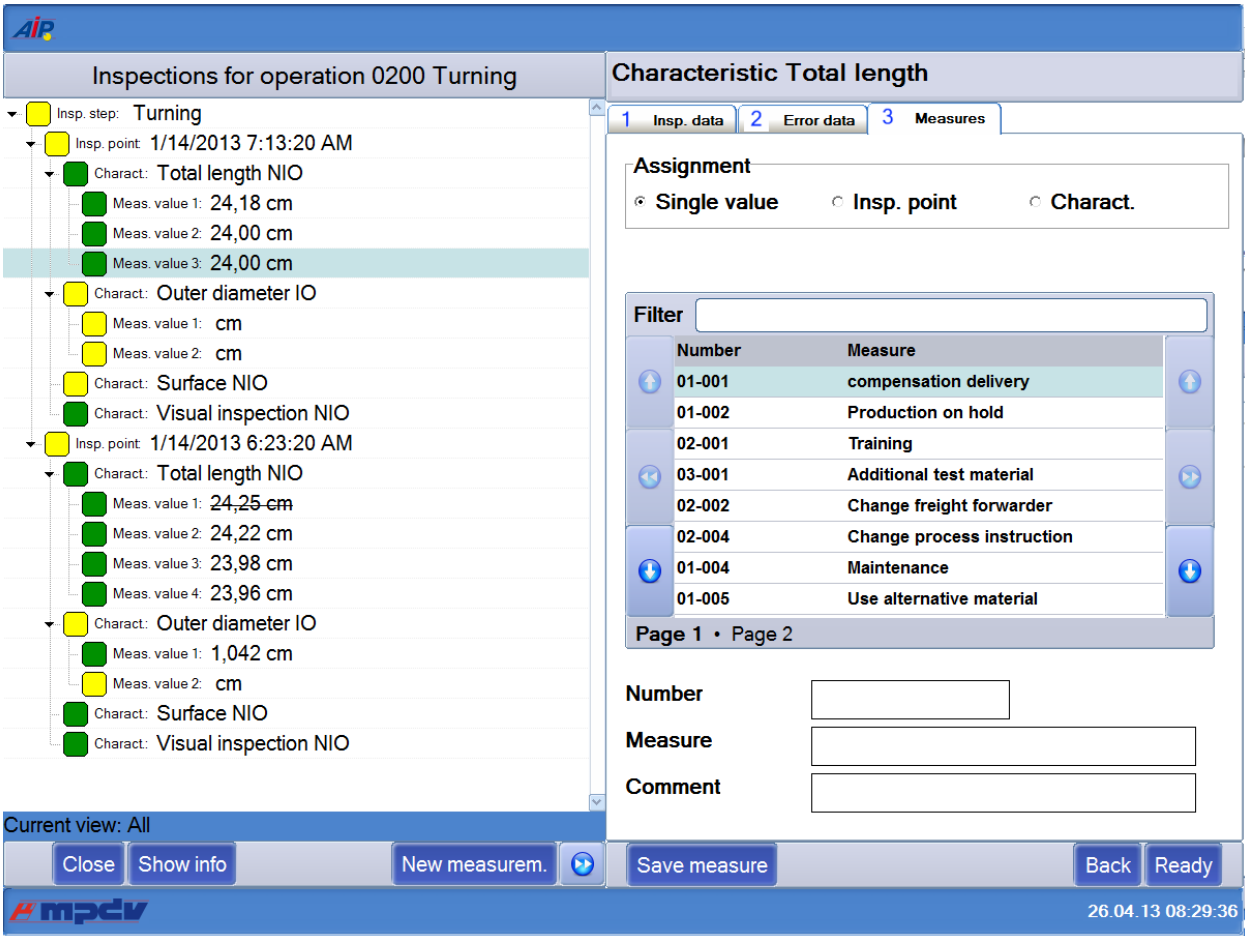
- Click the button *Save failure* to save the current failure entry. In this case, the user remains in the current index tab and can save additional failure analysis criteria.

- If you have not saved a failure and you click the button *Next* or *Back*, the program will automatically send a confirmation prompt.

- After clicking the button *Done*, the application saves the selected failure. You can now switch to another action element.

1.4.2 Recording of measures

The input dialog *Classic recording of measures* is integrated in many different input functions.



- If an *analysis selection catalog* is defined for a characteristic, this catalog specifies the possible entries.

- Enter a filter to restrict the measures displayed for selection.

Performance and operation: *“Classic recording of measures“:*

- Click the button ”save measure” to save the current measure. In this case, the user remains in the current tab and can save further measures

- The program displays a prompt if the button “Next“ or “Back“ is clicked without having saved the measure

- After clicking the button *Done*, the selected measure is saved. You can now switch to another action element.

- You can edit the text in the input field *“measure“.*

1.4.3 Mandatory recording of failures and measures (as of CAQ 8.2 + add-on)

If this function is enabled, you can be forced to enter failure data and/or measures when you record measured values.

To identify if a failure occurred, the following types are available:

- Automatically generated failure in HYDRA
- Quality status of the measured value

You can define the following failure data:

- Failure type
- Failure location
- Failure cause

Configure general settings in the file caq_dc_t.ini. You can define a separate configuration for each input type in the dialog-specific INI file.

1.4.4 Calculated characteristics

The input panel for *calculated characteristics* is always read-only.

You must update the display manually to show the results of the *calculated characteristics* in the *inspection list* when you have recorded the inspection results of all source characteristics.

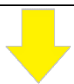
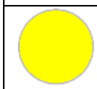
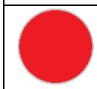
To this end, click the button “Update display”. 

The terminal must be ”online“ to request *calculated characteristics.*

☞ **Note**	**Individually calculated *characteristics*** The function to calculate characteristics individually is not supported.

1.4.5 Display and identification of the quality status (as of CAQ 8.2 + add-on)

Install the add-on to show the quality status of the measured value in the inspection list. The terminal calculates the quality status when the measured value is entered. Calculations include the tolerance and action limits and/or the quantities accepted and rejected. The following symbols are available:

	Upper tolerance limit violated (fail)
	Upper action limit violated (conditionally pass)
	Lower tolerance limit violated (fail)
	Lower action limit violated (conditionally pass)
	Number of non-conforming units (conditionally pass) (Non-conforming units < rejection quantity) and (Non-conforming units > acceptance quantity)
	Number of non-conforming units (fail) (Rejection quantity <= number of non-conforming units)

1.4.6 Automatic completion of inspection points (as of CAQ 8.2 + add-on)

If configured accordingly, you can use the add-on to complete inspection points automatically. The HYDRA server takes the usage decision (pass, fail) if operated in this mode. You can configure the following:

- All
  The system completes the inspection point automatically when the last measured value has been collected.
- Only valid measured values
  The system only completes inspection points with quality status "pass" or "conditionally pass".

When the inspection point is completed, the system continues processing (if auto navigation is enabled) or completes the inspection data collection.

1.4.7 Assigning a test equipment number

After installation of feature pack FP04-2018 (CAQ 8.2 add-ons), the function of the automatic assignment of a test equipment used in an inspection to the measured value or the attributive inspection is available.

The test equipment used (the test equipment number) is displayed in the MOC application *Single values* in field *Test equipment (used).* This field is contained in the list of available columns of the list *Single values.*

Condition for the automatic assignment of the test equipment used for an inspection: you must specify a test equipment number or a test equipment group for the relevant inspection plan characteristic. If you have specified a test equipment number for the inspection plan characteristic and therefore also the inspection step characteristic, this number is stored with the measured value recorded or the attributive inspection result for each inspection. If you have specified a test equipment group for the inspection plan characteristic and therefore also for the inspection step characteristic, the system identifies and stores the test equipment number as follows:

1. The system identifies the workplace where the measured value or the attributive inspection has been recorded. The required workplace number is derived from the respective sample.
2. The system uses the workplace number to identify the resource of type "PRM" (test equipment) in the resource list, which belongs to a test equipment group (resource family) that matches the one of the inspection characteristic.
3. If the system identifies more than one test equipment belonging to a matching test equipment group in the resource list, the first test equipment group identified for the measured value or the attributive inspection is saved.

In case of calculated characteristics including calculation of eigenvalue, the test equipment is only saved for the calculated value.

You cannot change the automatically identified and saved test equipment number using the editing function of the MOC application *Single values*.

1.5 Data collection with inspection points

➔ Inspection with inspection point

1.5.1 Inspection step

The input function "*inspection step"* includes only one tab. You use this function to display data because there is no interaction with the user.

The function shows information about the order, the operation, the article and the article name. For the QM, the function shows the inspection requirement and the inspection step.

The dialog does not allow interaction with the user (buttons, fields, etc.).

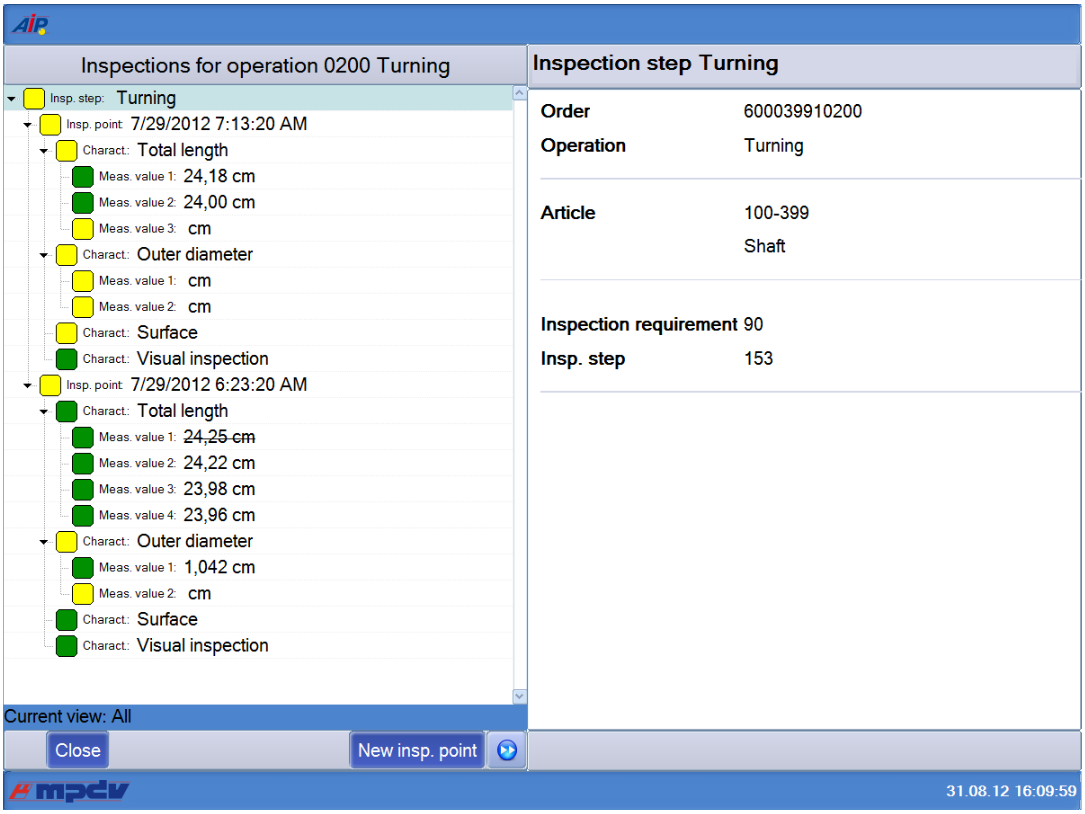


Figure: Function "inspection step" (right)

1.5.2 Inspection point

You **must not** change the structure of **workflows** and **dialog** settings of the data collection for *"inspection points".*

Discuss change requests with **MPDV Consulting** and/or MPDV CAQ Software Development.

The input function "*inspection point"* is made up of two index tabs.

By default, the first index tab *Identification* includes several fields. Via customization, you can specify if the fields are visible or invisible and if the fields are optional or mandatory fields.

By default, the second index tab "Details" also includes several fields. Also in the second tab, you can specify via customization if the fields are visible or invisible and if the fields are optional or mandatory fields.

Note the following for this input function:

- When the AIP terminal generates a new inspection point, the application completes the fields with default data:

- *Shop floor workstation* (machine) → current machine where the inspection results are collected
- *Date* (USER_D1) → current system date
- *Time* (USER_T1) → current system time

- The field *shop floor workstation* is read-only
- If the inspection point is not completed, the usage decision is set to *undefined* by default. In this case, the fields *Group* and *Code* are both left empty.

☞ **Note**	**Usage decisions** The usage decisions that are available for an inspection step are specified via a filtering of the catalog for this inspection step.

- If a usage decision other than *undefined* is selected, the inspection point is *completed* by clicking the button *Done* in tab *Details*. The inspection point now has the processing status *Completed*.
- If you do not want that the user takes usage decisions, you stop access to the selection list *Usage decisions* and to the fields *Group* and *Code.*
- The fields *Partial batch* and *ERP Batch* in tab *Details* become mandatory fields if configured accordingly.
- The key fields of the *"Identification"* index tab are only shown if the respective control flags have a numeric value greater than 0 and less than 99 *(mandatory fields).*
- Key fields are *never* shown if the values are less than or equal to 0.
- If values greater than or equal to 99 are entered, the key fields become *optional fields.* You can use a parameter to specify if apart from the *mandatory fields* of an inspection point, the *optional fields* are also displayed.
  --> see chapter Configuration qee_insppoint.ini

☞ **Note**

Failures / measures

If you do not use the QMS environment, the option to record failures or measures for the inspection point is not provided.

In a non-QMS environment, the recording of failures is integrated on the respective levels:

- Inspection requirement
- Inspection step
- Inspection point characteristic
- Sample
- Single values

1.5.2.1 Configuration qee_insppoint.ini

Entry	Comment
Section `[CONFIGURATION]`	

`SHOW_OPTIONAL_USERFIELDS=[ON,OFF]`	Option to show or hide optional fields of an inspection point.
	
	Default → `OFF`
	
	Example 1
	`SHOW_OPTIONAL_USERFIELDS=ON`
	Using this configuration, the terminal does not only display the mandatory fields of an inspection point (values ranging between 0-98), but also all optional fields (value greater than 98).
	
	Example 2
	`SHOW_OPTIONAL_USERFIELDS=OFF`
	The terminal displays only the mandatory fields (values ranging between 0-98) of an inspection point.

1.5.3 Inspection point characteristic

This type of data collection includes a single-part inspection. On the level of the characteristic, the terminal only shows a summary of the characteristic inspections for this inspection point.

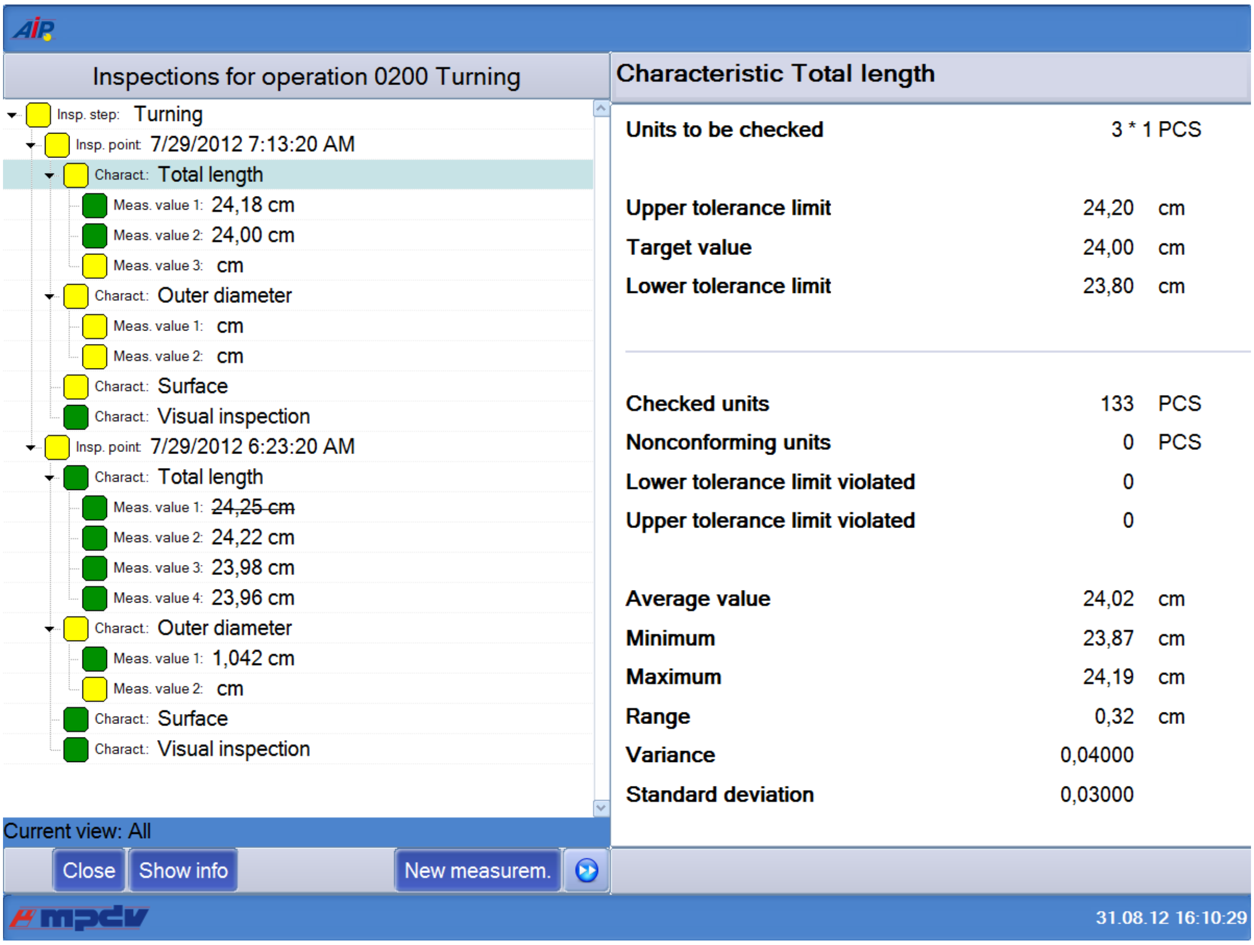


Figure: Input function "Inspection point characteristic"/ "inspection data" index tab

By default, this input function does not have any other dialogs (as *Classic recording of failures* for example).

The dialog does <u>not</u> allow interaction with the user (buttons, fields, etc.).

1.5.4 Attributive data collection

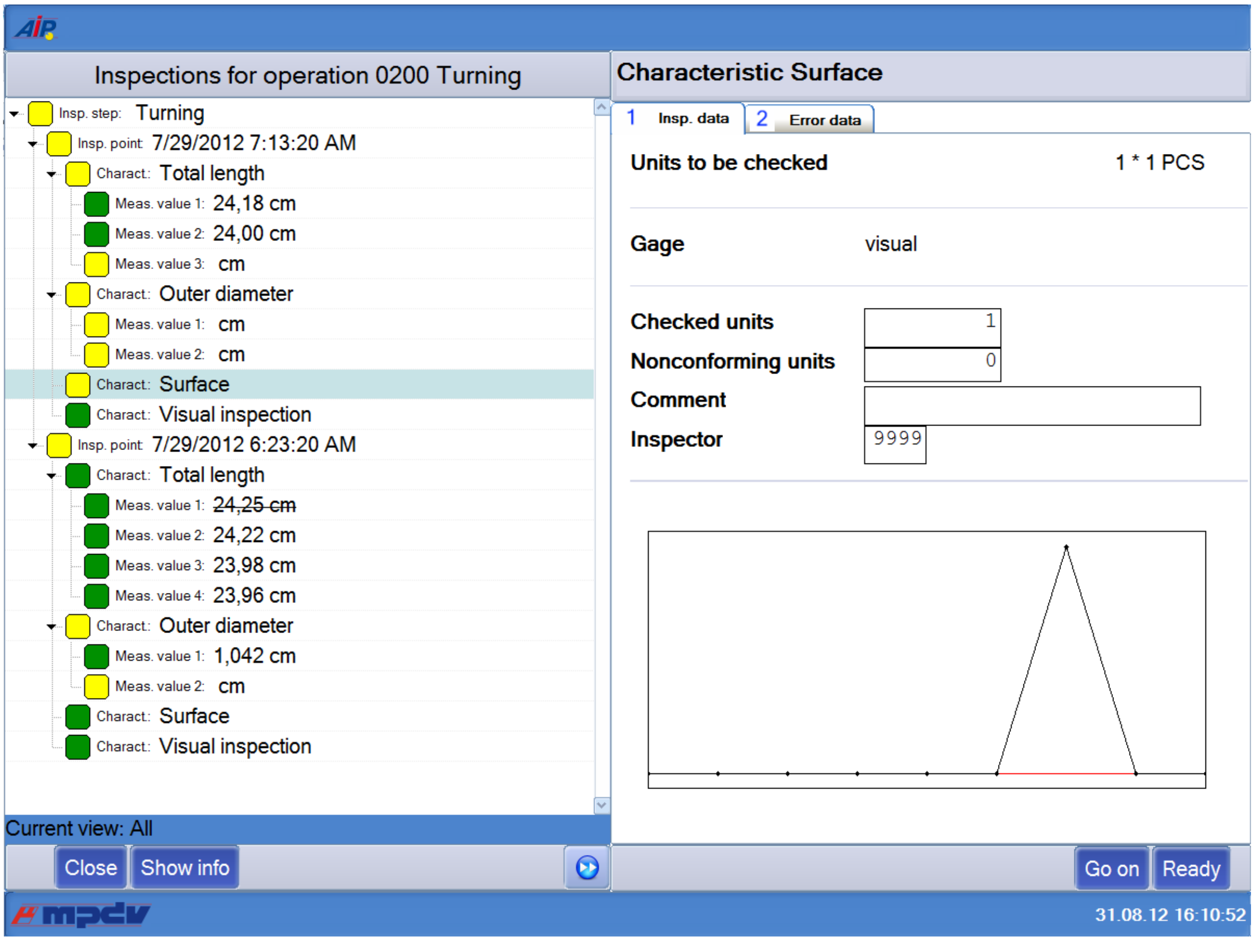


Figure: Input function "Attributive data collection / "inspection data" index tab

- By default, the value 0 is preset in the field *non-conforming units*.
- By default, the sample size is preset in the field *Checked units.* If the sample size is smaller than or unequal to 1, the field remains empty.
- The focus is initially set on the field *non-conforming units.*
- You can click the *Done* button to save the data entered for the *attributive characteristic.*
- If you click the *Next* button, the data entered for the attributive characteristic is saved.

☞
Note

Save data

To save the data entered, you can click the *Done* button or the *Next* button (→ go to next index tab). In both cases, the data is saved.

☞
Note

Number of checked units/ number of non-conforming units

The number of *non-conforming units* must be less than or equal to the number entered for *Checked units*.

☞
Configuration

Relevant configuration files in the "Inspection data" index tab

- `mm_be_st_pp_si.ini`
- `qee_mm_be_st_pp_si.ini`

The recording of attributive characteristics in tab *Failure data* is additionally provided. This is the *Classic recording of failures* on the AIP terminal. This dialog is integrated in many data collection functions.

In this input dialog, you record the failure data of the sample for the respective inspection point or characteristic.

☞
Note

Classic failure recording/ failure data

By default, you can record the following failures:

- Failure types
- Failure locations
- Failure causes

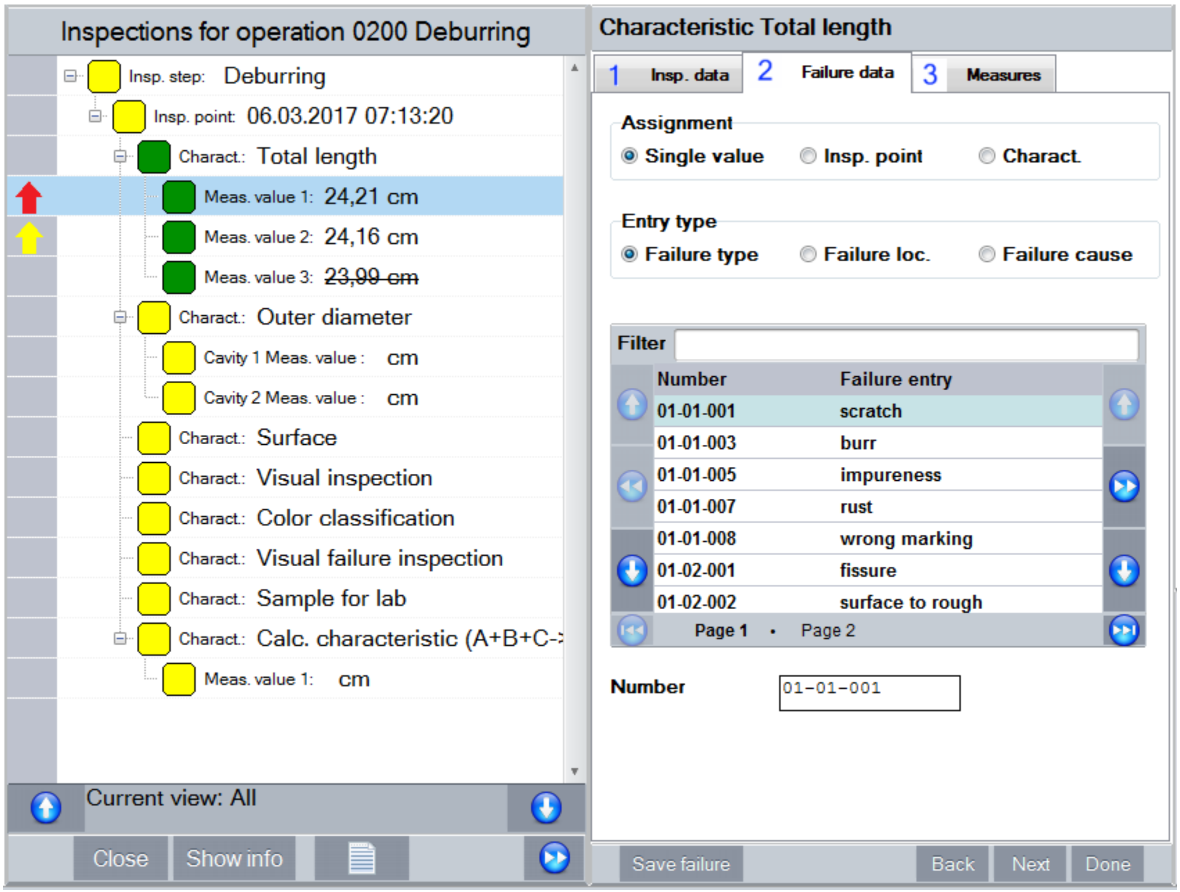


Figure: Input function "Attributive collection / "failure data" index tab

☞
Configuration

Relevant configuration files in tab "Failure data"

- `ctaiplay.ini`
- `mm_be_st_pp_si.ini`
- `qee_err_classic.ini`

1.5.5 Variable collection

You record the *single values* for the *sample* of the *inspection point*.

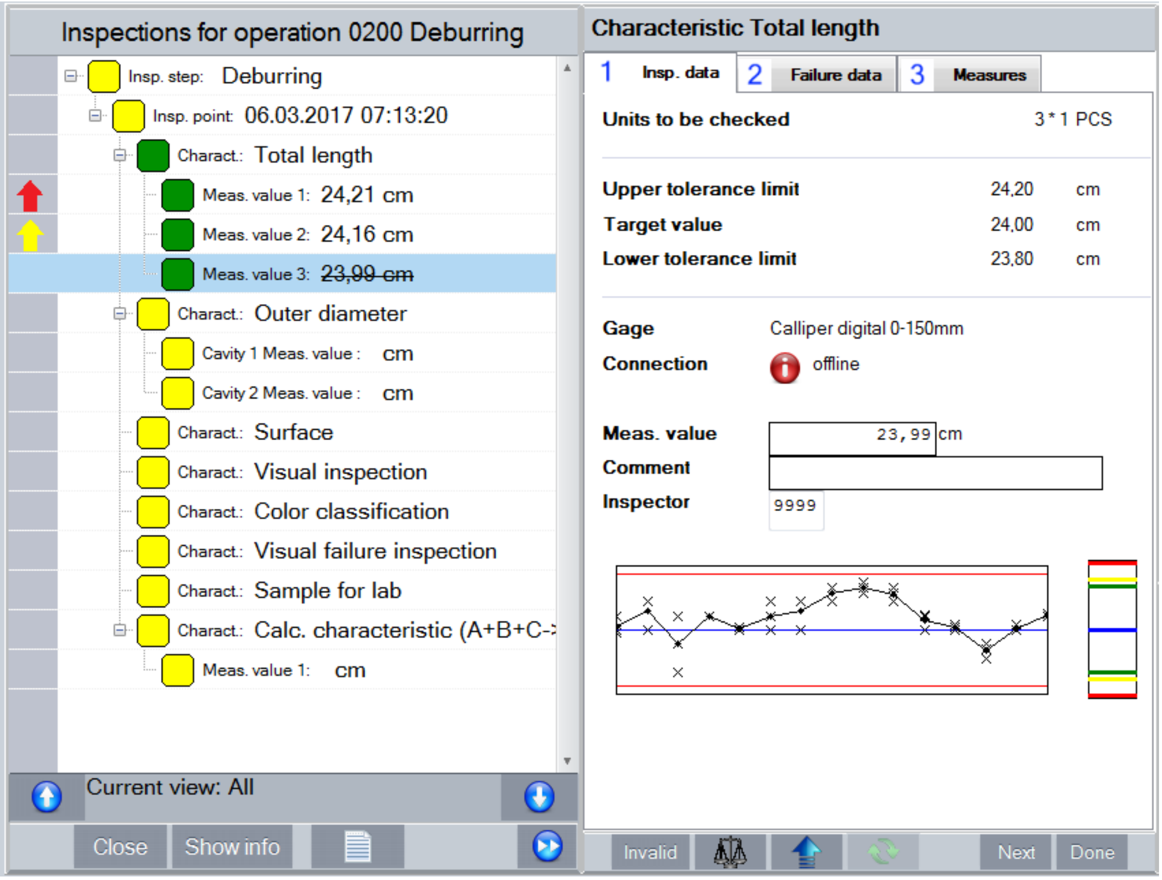


Figure: Input function "Variable recording / "inspection data" index tab

If the MDI connection is active, the system requests all MDI driver values of the corresponding channel without setting filter criteria.

1.5.6 Inspection chart

Use an *inspection chart* to record different *failure types* concerning a *sample* relating to an inspection point.

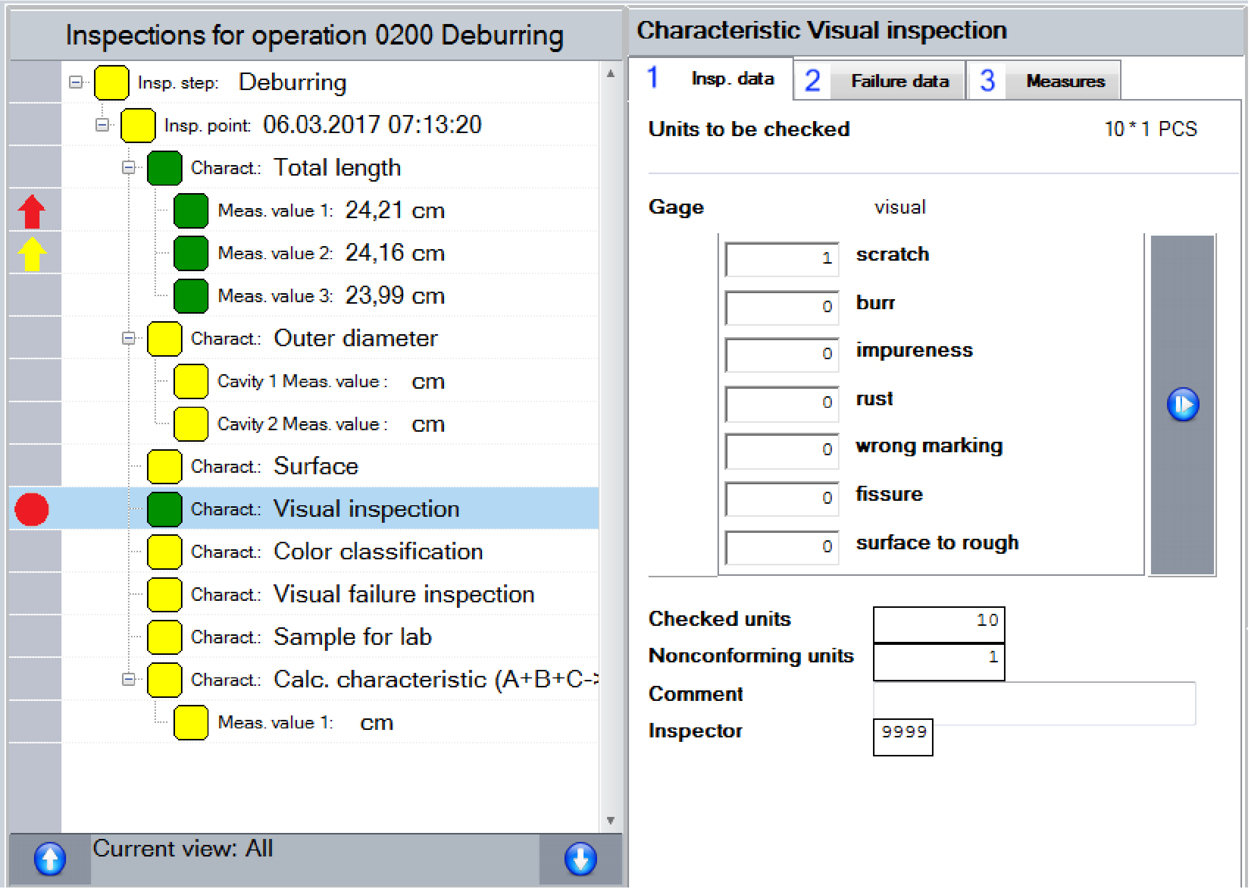


Figure: Input function "Inspection chart"/ "inspection data" index tab

The selection of available *failure types* depends on whether an *Analysis selection catalog* has been assigned to the characteristic or not.

If an *Analysis selection catalog* exists, the catalog is used to populate the entries of the inspection chart. The advantage is that there is an individual and in some cases a small selection.

If no analysis selection catalog has been assigned to the characteristic, all existing failure types are available in the inspection chart.

- The default value 0 is preset in the fields *Non-conforming units* and for each (failure type) entry of the inspection chart.
- By default, the sample size is preset in the field *Checked units.* If the sample size is smaller than or unequal to 1, the field remains empty.
- Initially, the field *Checked units* has the focus.
- Save the inspection chart by clicking the *Done* button.

- Save the inspection chart by clicking the *Next* button.

☞ **Note**

Save data

To save the data entered, you can click the *Done* button or the *Next* button (→ go to next index tab). In both cases, the data is saved.

- The system automatically calculates the number of *non-conforming units* from the *sum total of failures entered in the inspection chart*. At this point, the user can manually overwrite the automatically calculated number of *non-conforming units*.

☞ **Note**

There are no automatic failures in the inspection chart

The system does not generate automatic failures for inspection charts.

☞ **Note**

Number of checked units/ number of non-conforming units

The number of *non-conforming units* must be less than or equal to the number entered for *Checked units*.

☞ **Configuration**

Relevant configuration files in the "Inspection data" index tab

- `ctaiplay.ini`
- `mm_be_st_pp_fs.ini`
- `qee_mm_be_st_pp_fs.ini`

For the data collection in the inspection chart, the terminal also provides the tab *Failure data*. This is the *Classic recording of failures* on the AIP terminal. This dialog is integrated in many data collection functions.

Of particular interest in the context of the *Inspection chart* is that the *failure data* is recorded for each sample of the relevant inspection point.

☞ **Note**

Classic failure recording/ failure data

You cannot record *failure types* here.

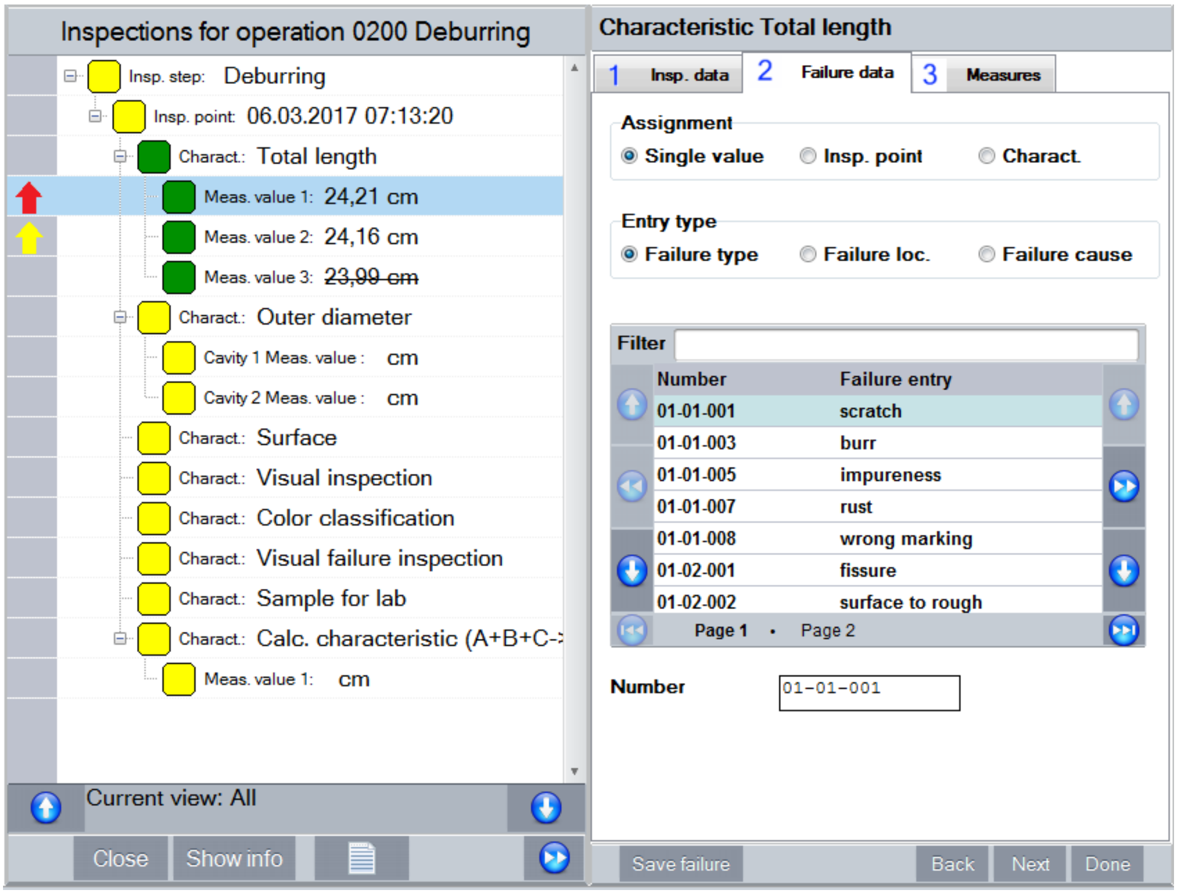


Figure: Input function "Inspection chart"/ "failure data" index tab

Configuration	Relevant configuration files in tab "Failure data"
	- `ctaiplay.ini` - `mm_be_st_pp_fs.ini` - `qee_err_classic.ini`

1.5.7 Data collection for cavities

You must generate inspection points to collect data for cavities because the tool is assigned to the inspection point and the tool specifies the number of cavities. When the inspection point is created, the system automatically uses the tool assigned to the operation for the inspection point.

If the function extension for the in-production inspection is not available, the collection for cavities is limited to variable characteristics with inspections based on characteristics. If the function extension for the in-production inspection is available, the collection for cavities also covers attibutive characteristics and supports a data collection with reference to pieces.

In the inspection point, the tool number must not be greater than 15 digits. If the tool assigned to the operation has more digits, you must create a custom dialog for the terminal or the terminal group. You can use the dialog "QEE_INSSPOINT" to change the field length.

You record the *single values* for the *sample* of the *inspection point*. The example below is based on a collection of measured values for cavities that is based on characteristics.

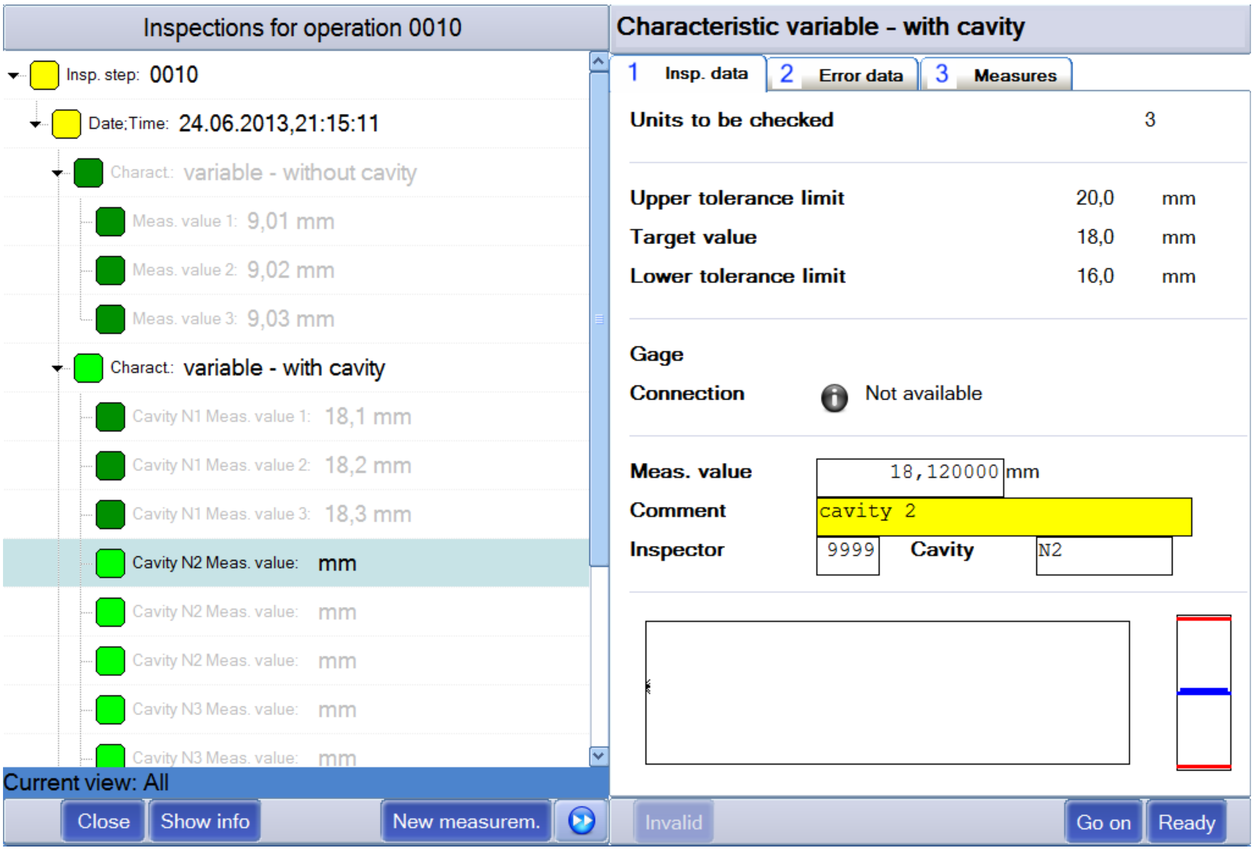


Figure: Input function “Variable recording including cavity“ / tab “inspection data“

- Initially the field *measured value* has the focus.

- In the *Cavity* field, the system enters the value that is specified for the *Tool* in the *cavity list* (available primary tool of the logged on operation). You can change this field value, as long as the measured value is not saved.

☞
Note

Manual input of a cavity

If you manually enter a value in the *Cavity* field, no validation check is run that checks if the cavity entered actually exists. However, you must enter a value in the *Cavity* field (mandatory field).

- Click the *Done* button to save the recorded data of the *variable characteristic with cavity.*
- The C*avity* field is read-only and can no longer be edited when you have entered a measured value.
- Click the *invalid* button to invalidate the recorded value. Click the button *new measurement* if you want to replace the invalid value with a new value.
- Click the *Next* button to save the recorded data of the *variable characteristic with cavity.* Use this function if you want to collect further data in the workflow, e.g. *failure recording*
- The sample size of the characteristic is calculated via the product of *sample size * number of cavities* included in the *cavity list.* Consequently, the sample size of the inspection plan characteristic matches the number of measured values to be recorded for each cavity. Empty rows for measured values are displayed, once the inspection list has been updated or opened the next time.

The sample size for cavity-related data collection is not subject to restrictions. Consequently, validation checks are not performed. This means: with cavity characteristics, the inspection scope indicator is always set to "EGAL". The color of the node point therefore does not change and is light green.

☞
Note

Terminal behavior: Inspection results of different machines for a cavity

It is possible that you record different samples for a cavity of an (inspection step) characteristic. For example, this is the case if the inspection results are from different machines.
The terminal does not show the results of both machines in the inspection list. Only the results of the current machine are shown.

☞ **Note**

Display of measured values in the inspection list

You cannot configure the label for cavity-related measured values.

☞ **Note**

Re-sorting of measured values in the inspection list

The application only re-sorts the measured values of a characteristic that relates to cavities
a) when you call the *recording of inspection results* the next time

or

b) when you manually update the *values displayed.*

→ See section Cavity list

In case of an attributive inspection for a cavity, the cavity number is only displayed in the inspection list. The cavity number is not displayed in the actual input dialog.

In case of a cavity-related data collection for a variable characteristic with piece-related inspection, you can define the content that is displayed in the inspection list for the node "shot/part". You make this definition in the configuration file "caq_dc_t.ini" (AIP sub folder functions) in section "[GROUP_BY_SP_NEST_SCHUSS]". In section "[MW_SCHUSS_NEST]", you configure the content that is displayed for the node of the measured values in the context of a piece-related inspection. For details on the configurations, refer to the document "Configuration_AIP-QM.pdf".

1.5.7.1 Tool

☞ **Note**

Requirements for the operation: resource type

You must define a resource of the type `WNR` for the operation in order to use the cavity function with the AIP terminal.

If required, you can change the *tool* while working with an inspection point. You can do so in the *"inspection point"* input function of the terminal.

Please note in this context that the cavity list (contents and/or order) might change if you change the tool.

Changing the tool of an inspection point when the terminal is in *"offline"* status *does* *not* *affect* the currently available cavities. Updating the cavity list requires the *"online"* status.

→ See section Cavity list

1.5.7.2 Cavity list

The *cavity list* results from the assignment of cavities to the tool.

The terminal shows the *cavity list* according to the order of the inspection list defined in the MOC.

The below changes can affect the content of a *cavity list:*

- Assignment of another tool to the inspection point
- Blocking of (individual) cavities

  Example:

  If a cavity is blocked (e.g. in MOC) during a terminal session or the tool is changed directly on the terminal, this cavity is removed from the terminal, once the inspection list has been updated. Provided that measured values have already been recorded for this cavity, they are still shown in the inspection list. If required, the inspection list shows empty rows for measured values in order to reach the sample size.

☞ **Note**

Modification of the cavity list on the terminal

The terminal uses information from cavity assignments available at the time when the cavity list is requested (*not* at the time of generating the inspection point).

The inspection list shows the changes made, once you have clicked the button

.

You can make such changes directly on the terminal or the MOC.

→ See section Tool

1.5.8 Visual assignment of failures (as of CAQ 8.2)

Input type `BEWERT_STICHPR_PPUNKT_RASTER`

Use this input type to enter the defect positions in the grid.

The system only supports this input type in combination with inspection points.

You must not change the dynamic dialog `QEE_MM_BE_ST_PP_RA` without contacting MPDV.

This dialog shows the characteristic document (not the *inspection requirement document*) at position 1. It must be of the type "FILE".

The picture size might have an impact on the time it takes to display the picture in the workflow. Therefore, you should use small-size pictures.

The AIP automatically scales the pictures to the ideal size.

We recommend to use square pictures.

The following formats are supported:

- JPEG, JPG, PNG

In order to be able to divide a graphic into different areas, you must define a pattern for the x-axis and y-axis of the available inspection plan characteristic.

Use a comma to separate the data entered for the inspection plan characteristic.

But the comma must neither be the first nor last character in an axis definition. Several commas must not succeed one another.

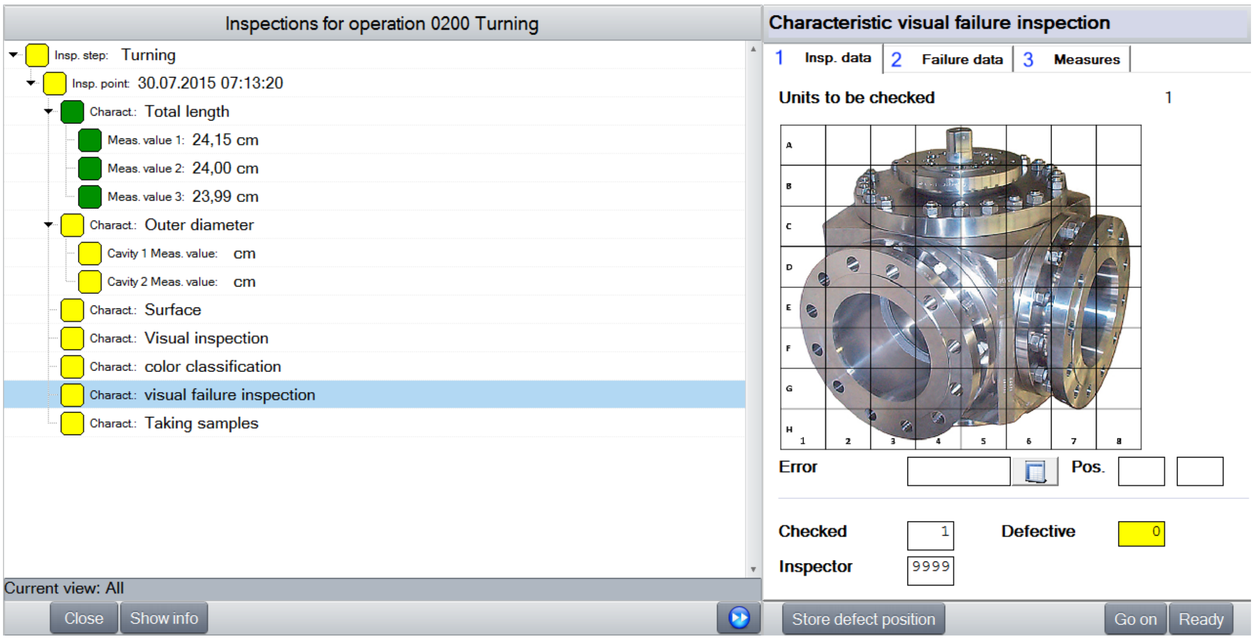


Figure: Visual defects recording

- Select a "failure type" to complete the field "failure". Click the button behind the "failure" field to open a list of "failure types" where you can select a failure. The respective *number* of the failure type is then entered in the "failure" field.

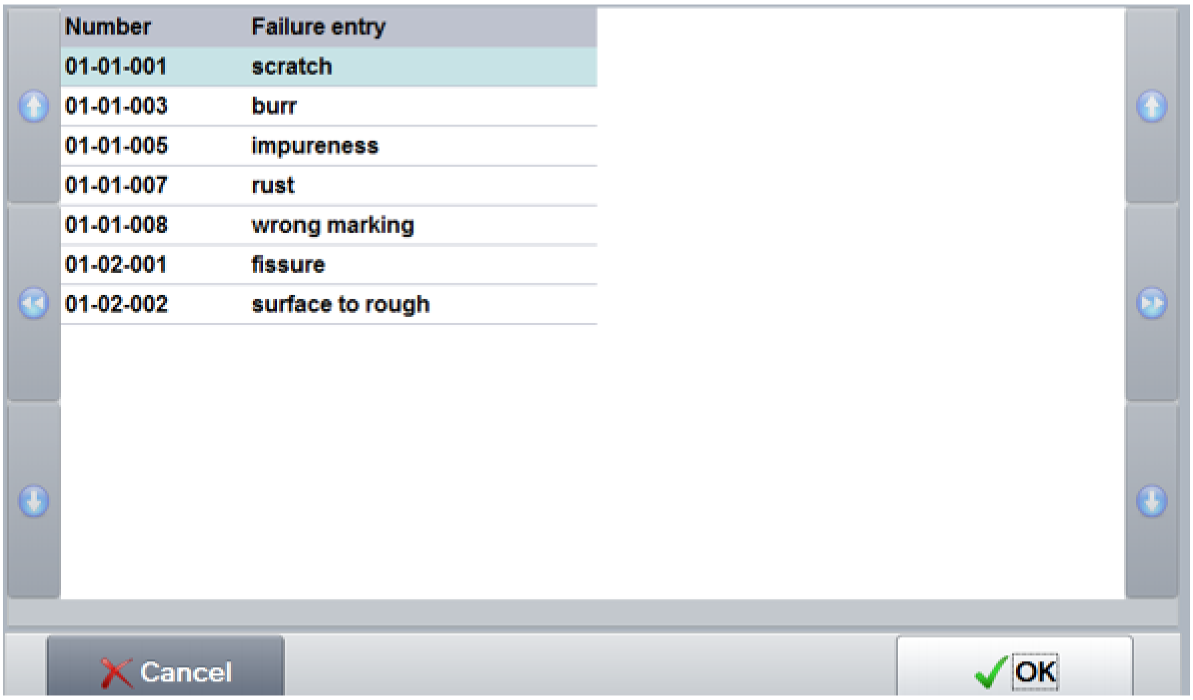


Figure: Selection list of failure types

- The "position" fields are completed with the coordinates of the area you selected in the graphic.

- By default, the sample size is preset in the field *Checked.* If the sample size is smaller than or unequal to 1, the field remains empty.

- The default value 0 is preset in the field *Defective.*

☞
Note

Automatic assignment of the field *Defective*

To optimize data collection, the application assigns the value 1 in the field "defective" when the first defect position is entered if the field is empty or includes the value 0.

For all further actions, this automatism does not apply.

- Initially, the field *Failure* has the focus.

Click the button "save defect position" to save the following values:
- the selected failure type,
- the specified position and
- the inspection result (contents of the fields "checked", "defective" and "inspector").
The system does not change to another characteristic. The characteristic is still selected so that you can enter further defect positions.

Saving *defect positions:*

Note	**Requirements for saving defect positions** An *inspection result for the characteristic* must be available. Only then, you can enter the position of failures.

- The *"failure"* field is mandatory.
- The fields where you enter the *position* are mandatory.
- You can save the collected data by clicking the button "*generate samples".*

Clicking the buttons "Done" or "Next" only saves the *inspection result of the characteristic* but not the defect position. Click the button "done" to change characteristics.

Saving the "inspection result of the characteristic":

- The "checked" field is mandatory, 0 is not a valid value.
- The "defective" field is mandatory, 0 is a valid value.
- The "inspector" field is mandatory.
- You can save the collected data by clicking the button "*Done".*
- You can save the collected data by clicking the button "*Next".*

Click the button "store defect position" and then "Done" to enter one defect position.

1.5.9 Inspections based on catalogs (as of CAQ 8.2)

Input type `CODE_STICHPR_PPUNKT_SIMPLE`

Use this input type to enter attributive characteristics via catalogs.

The system only supports this input type in combination with inspection points.

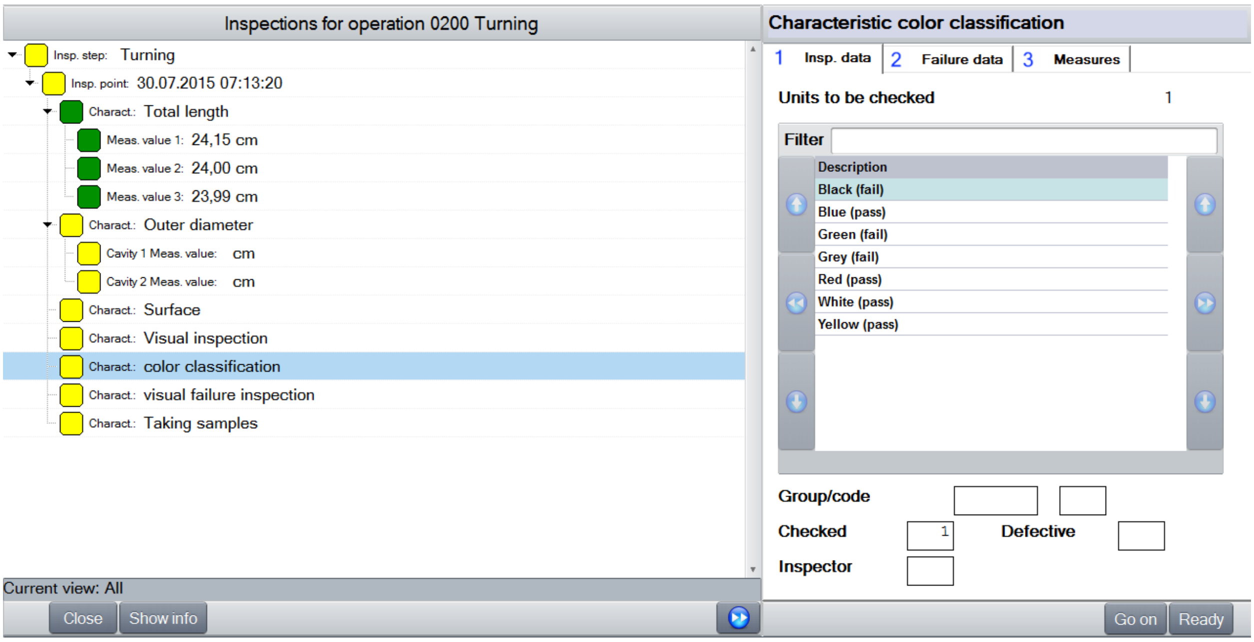


Figure: Inspections based on catalogs

- Select an entry from the above list to complete the fields *"group/code"*. Both fields are read-only.
- By default, the sample size is preset in the field *Checked.* If the sample size is smaller than or unequal to 1, the field remains empty.
- The default value 0 is preset in the field *Defective.*

☞ **Note**

Automatic assignment of the field *Defective*

To optimize data collection, the application enters the value "1" in the field *Defective* if you select an entry classified as "fail". Requirements: The field *Defective* is empty or it includes the value "0" and the inspection result has not yet been saved.

If these conditions are not met, the field is not subject to the automatism.

- Initially, the field *"defective"* has the focus.

Saving *inspection results:*

- The fields *"group/code"* are mandatory
- The "checked" field is mandatory, 0 is not a valid value.
- The "defective" field is mandatory, 0 is a valid value.
- The "inspector" field is mandatory.

- You can save the collected data by clicking the button "*Done*".
- You can save the collected data by clicking the button "*Next*".

1.5.10 Inspections based on catalogs, random selection (as of CAQ 8.2)

Input type `CODE_STICHPR_PPUNKT_ZUF_SIMPLE`

Use this input type to enter attributive characteristics via catalogs.

This input type is similar to the *catalog-based inspection* using the input type `CODE_STICHPR_PPUNKT_SIMPLE`. This input type provides a random selection of entries that you can select to complete the *group/code* field.

The system only supports this input type in combination with inspection points.

→ See section 1.5.9

1.6 Data collection without inspection points

→ Sample-related inspection

1.6.1 Attributive data collection

This function is equivalent to the input type `BEWERT_STICHPR_PPUNKT_SIMPLE`

→ see chapter Attributive collection

The data collected always refers to a (machine-dependent) *sample.*

The "Failure recording" dialog configured for this purpose records defects/failures relating to *samples.* In contrast to the input type `BEWERT_STICHPR_PPUNKT_SIMPLE` that records defects/failures relating to *inspection points*.

1.6.2 Variable collection

The display of data is similar to the display of the input type `MESSW_ESTCK_PPUNKT_SIMPLE`.
→ See section Variable collection

The data collected always refers to a (machine-dependent) *sample.*

The "Failure recording" dialog configured for this purpose records failures with reference to *samples.* This is different for the input type `MESSW_ESTCK_PPUNKT_SIMPLE` that records failures with reference to *inspection points*.

1.6.3 Inspection chart

The display of data is similar to the display of the input type `BEWERT_STICHPR_PPUNKT_FSK`.
→ See section Inspection chart

The functions are the same as for the input type `BEWERT_STICHPR_PPUNKT_FSK`.

The data collected always refers to a (machine-dependent) *sample.*

The "Failure recording" dialog configured for this purpose records failures with reference to *samples.* In contrast to the input type `BEWERT_STICHPR_PPUNKT_FSK` that records failures with reference to *inspection points*.

1.7 Sampling

1.7.1 Sampling (simplified)

The input function “sampling“ includes one single tab. This input function provides the user with the possibility to trigger the generation of samples.

By default, the *sampling* tab only shows the field *sample group.*

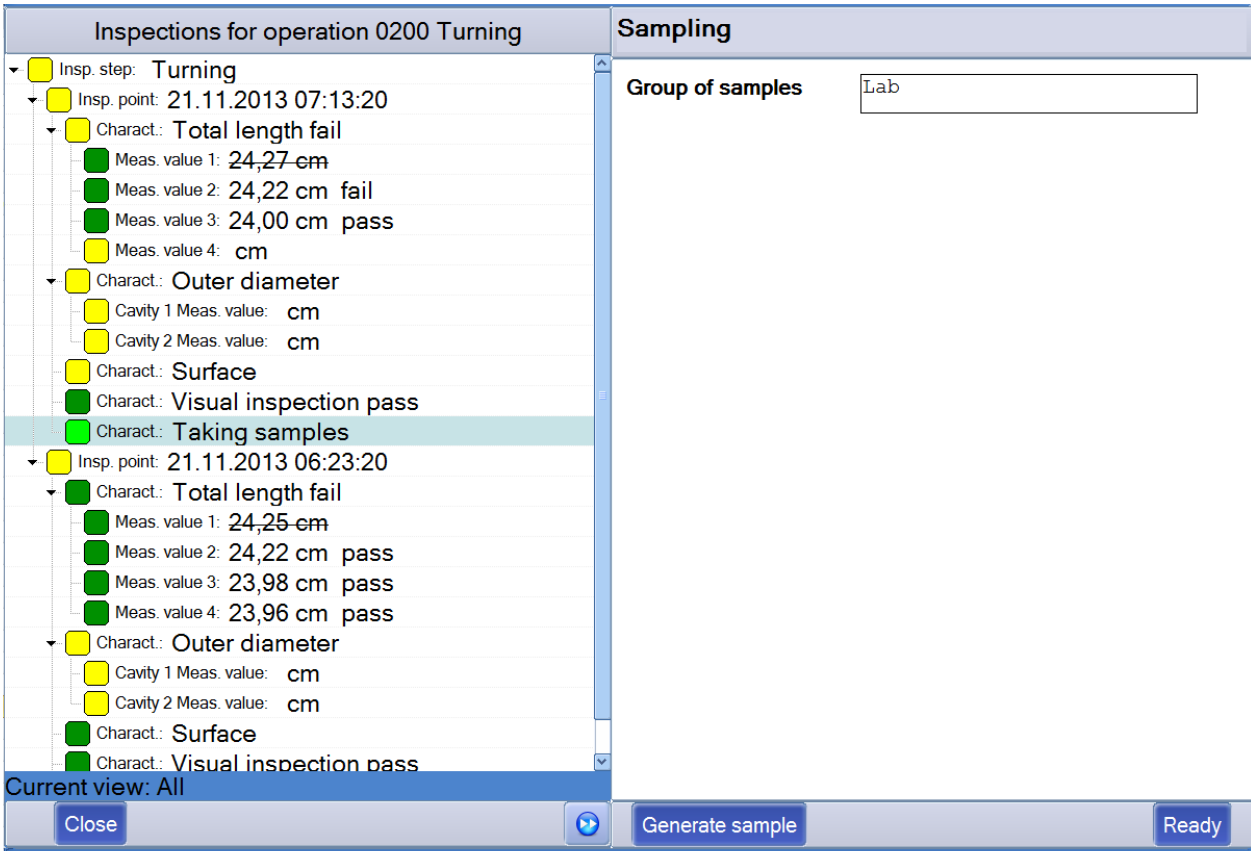


Figure: Input function “Sampling“ (right)

- Click the *generate sample* button to generate a new sample. If processing is successful, a message will appear showing the newly *generated sample number.*

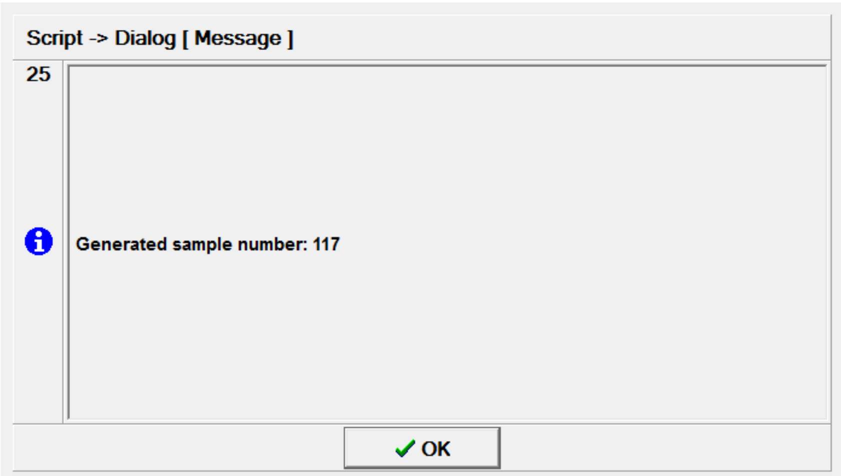


Figure: Message with the generated sample number

☞ **Note**	**Behavior in the event of failures** If an error occurs while generating the sample, the following message will appear: ➔ **Sample could not be generated** This message only occurs if the terminal is online.

- The collection status of the sampling characteristic is always constant and never changes. The collection status is centrally configured in the system and can have the following values:
  - Data collection possible
  - Data collection required (default setting)

  → see section Collection status

- If you click the *Done* button, no further action will take place. You can now manually switch to another action item. No actual inspection is performed for a sampling characteristic. For this reason, the default setting (collection required) remains unchanged even when the sample has been generated. For this reason, you must change the characteristic manually after generating the sample.

☞ **Note**	**Behavior in offline status** A message informs the user that the activity has been buffered. Once the terminal is again connected with the HYDRA server, all activities that have been buffered will be processed one after the other. You cannot generate samples in the offline status. Consequently, the user does not get a new sample number. **If samples are generated while the buffer is being processed (as described above), the terminal will not show the generated sample numbers for technical reasons.**

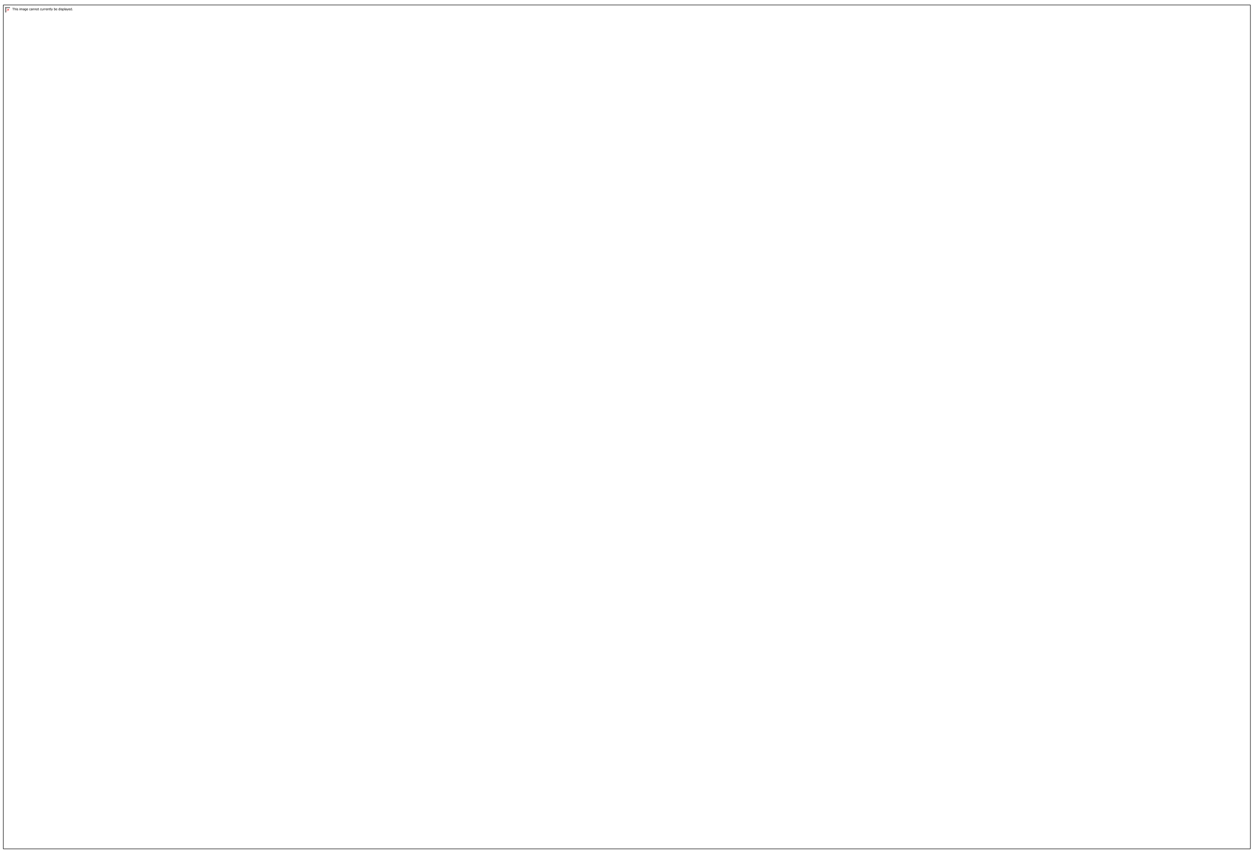


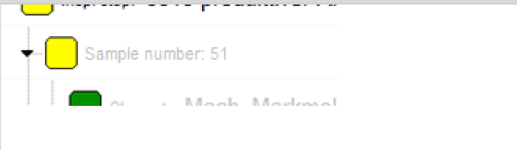
Figure: (Inspection station for samples) Example of a recording of data for an inspection point (for a sample) that is generated by sampling

☞
Configuration

Useful layout modifications in the inspection list at *inspection stations for samples*

Via the *generated sample number,* you can identify inspection points that have been generated when a sample has been created.

For this reason, it might be a useful layout modification to show the sample number in the *inspection list,* especially at *inspection stations for samples*.



➔ Please contact your MPDV CAQ consultant.

1.7.2 Advanced Sampling (as of CAQ 8.2)

The data collection function *"advanced sampling"* includes one single tab. This input function provides the user with the possibility to trigger the generation of samples.

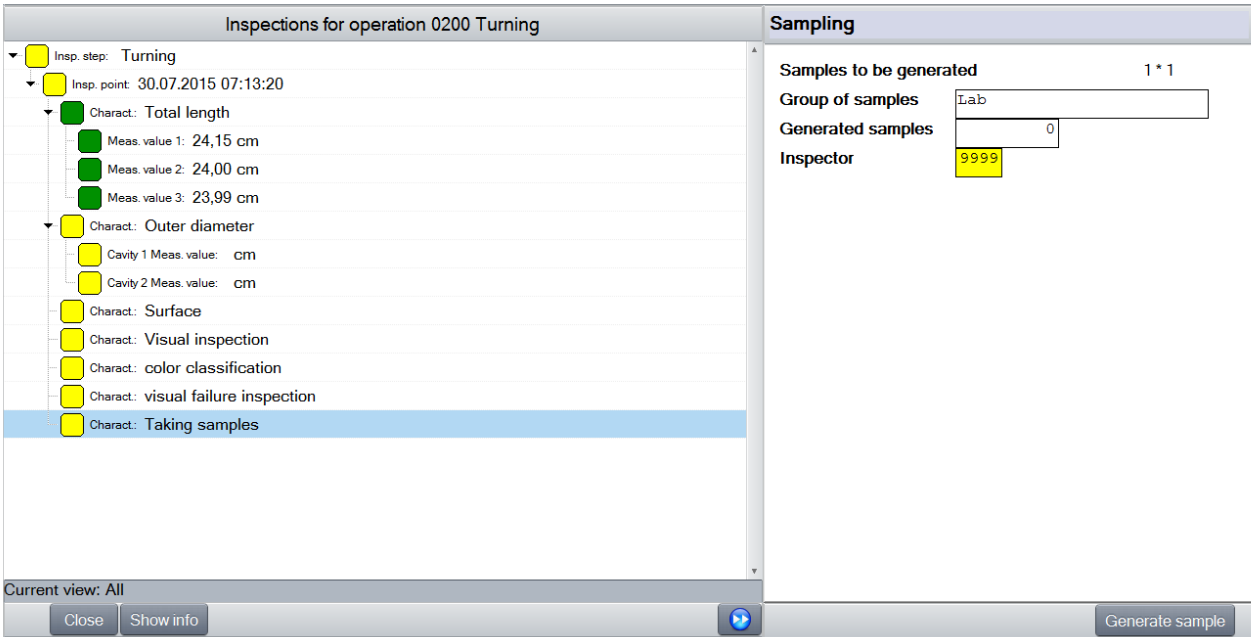


Figure: Data collection function "advanced sampling" (right)

- Click the *"generate sample"* button to generate a new sample. If processing is successful, a message appears indicating the new *"sample number".*

  Then, as it is also the case with other characteristic types, characteristics are changed and/or the next inspection point is "selected".

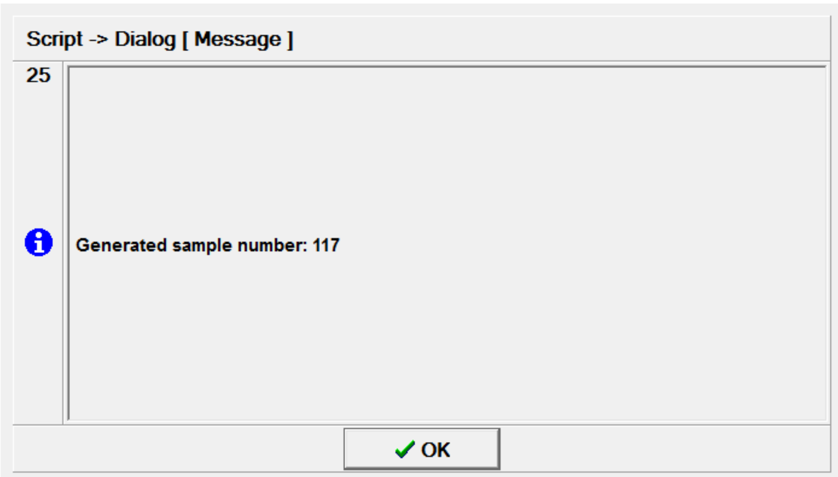


Figure: Message with the generated sample number

☞
Note

Behavior in the event of failures

If an error occurs while generating the sample, the following message will appear:

➔ **Sample could not be generated**

This message only occurs if the terminal is online.

- In contrast to the characteristic of a *simple sampling,* the data collection status for the *"advanced sampling characteristic"* is not constant. It is similar to that of *"attributive characteristics".*

  Other than for the attributive characteristic it is not the field *"checked units"* that is critical for changing the data collection status but the field *"generated samples".*

  The *"generated samples"* field informs the user about the number of generated samples.

  → see section Collection status

☞
Note

Behavior in offline status

Activities are buffered if the terminal is offline.

Once the terminal is again connected with the HYDRA server, all activities that have been buffered will be processed one after the other.

If samples are generated while the buffer is being processed (as described above), the terminal will not show the generated sample numbers for technical reasons.

☞
Note

Posting required

Irrespective of the defined sample size, you can generate more samples than specified for sampling characteristics. In this case, you can complete the inspection point using the "posting required" option.

1.8 Information on characteristics

Use the function *Information on characteristics* to show context sensitive information. Technically speaking, information on characteristics can be described as a workflow made up of index tabs that are shown and hidden as required.

Click the "Display info" button to request information on characteristics.

The subchapters that follow deal with the information provided by the terminal.

1.8.1 Description

This index tab shows the data relating to the current characteristic and the most important information about the corresponding operation and the defined test equipment/gage.

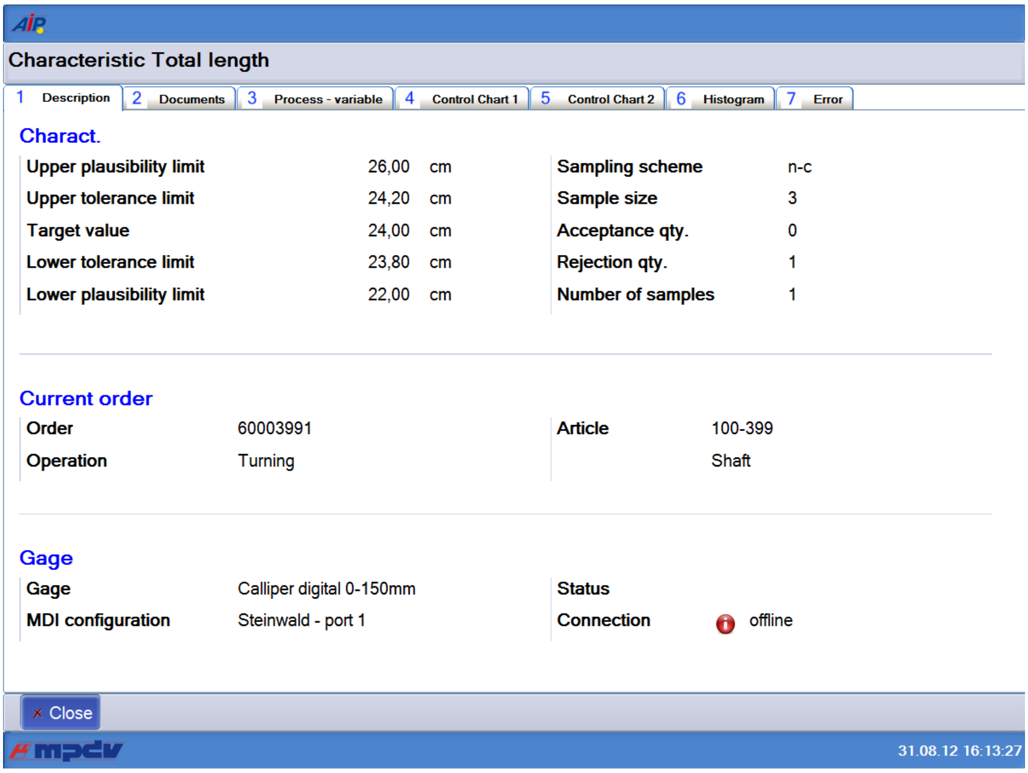


Figure: Index tab "Description" in information on characteristics

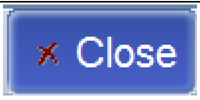
The information referring to the connection status of the test equipment/gage is derived from the corresponding MDI status after initialization. The following statuses are available for the connection status:

MDI test equipment connection is *online*

MDI test equipment connection is *offline*

No MDI test equipment connection is defined or can be identified and as such is *not available*

Note	The *Description* index tab is a permanent component of the *information on characteristics* function.

	Exiting *information on characteristics*

1.8.2 Documents

This index tab shows both documents defined for the inspection order characteristic and documents assigned to the inspection requirement.

The inspection requirement documents are shown in a first block (sorted by item number) and the characteristic documents in a second block (also sorted by item number).

As such, only those documents assigned to the option "display during inspection" are shown during the inspection process.

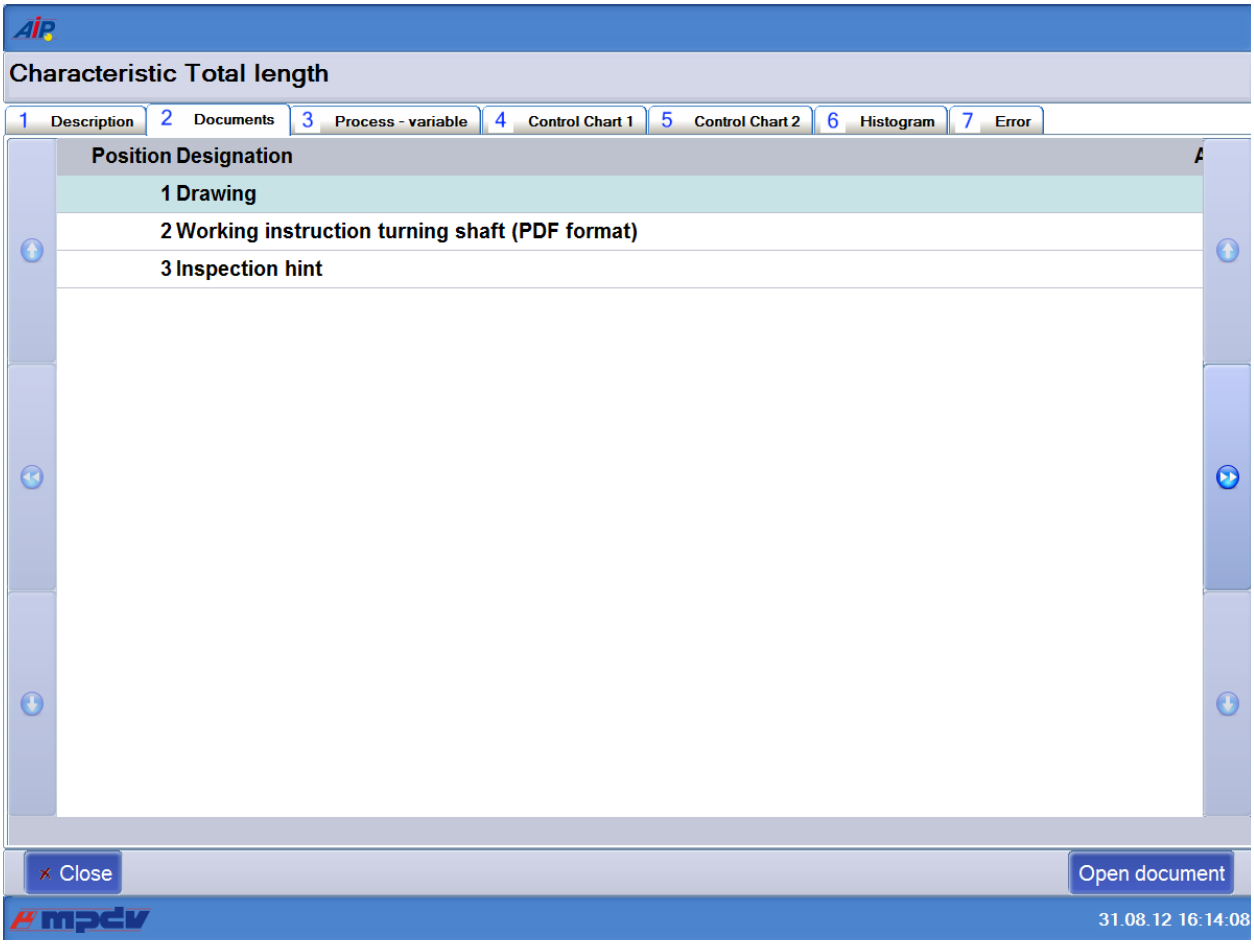


Figure: Index tab "Documents" in information on characteristics

	Exiting *information on characteristics*
	Calling the document for the selected entry in tab "Documents". Depending on the document, the system uses a different processing to display the document. → see below for more information.
	Button to close the document window.

If you click the button *Open document,* the system behavior is as follows for the respective document entry and AIP configuration:

The internal AIP display component displays the document.
In this case, a new screen opens displaying the content of the document. This new window takes up the entire area that the characteristic information currently takes up. So, except for this window, you can only see the AIP header and footer.
This window is shown as illustrated below:

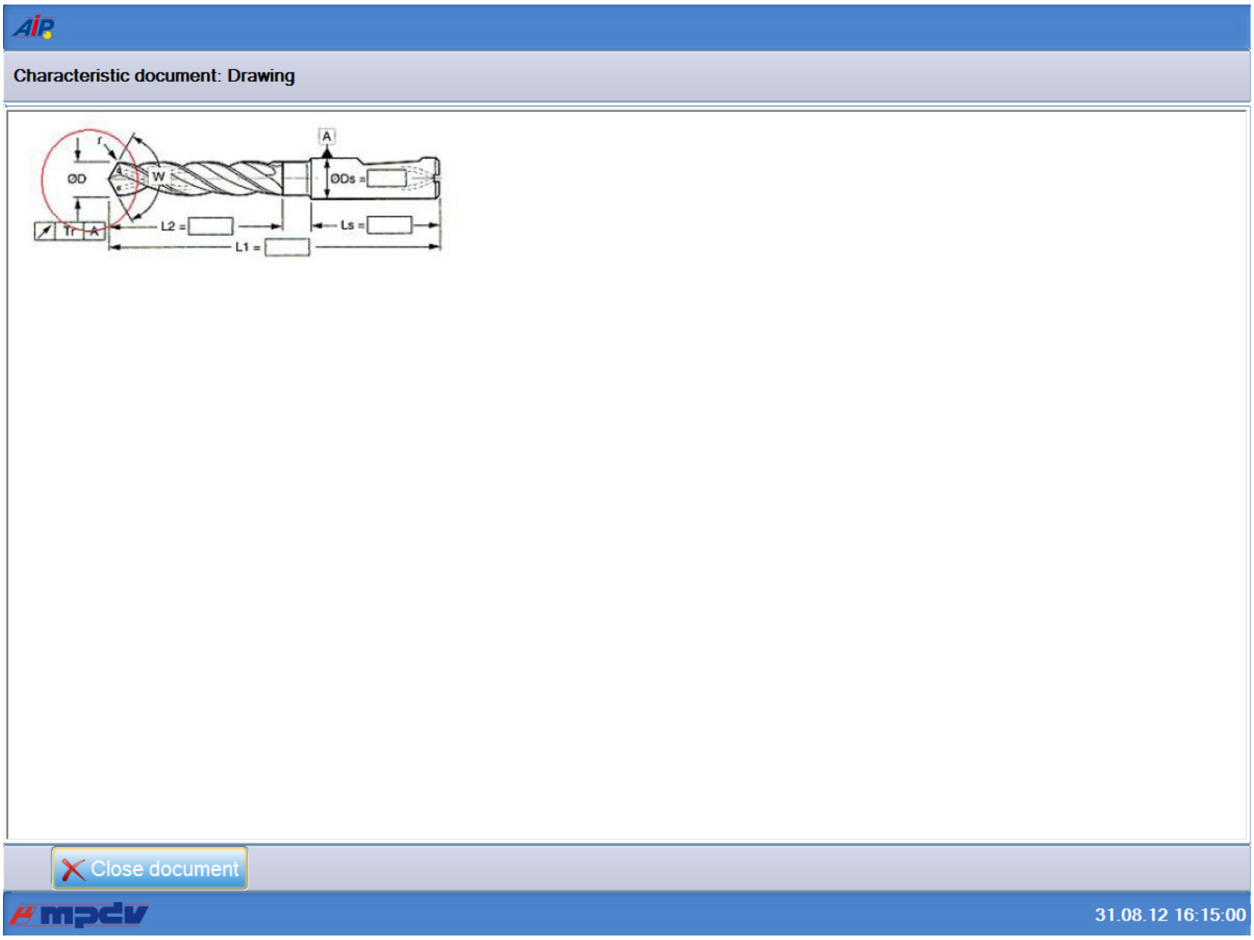


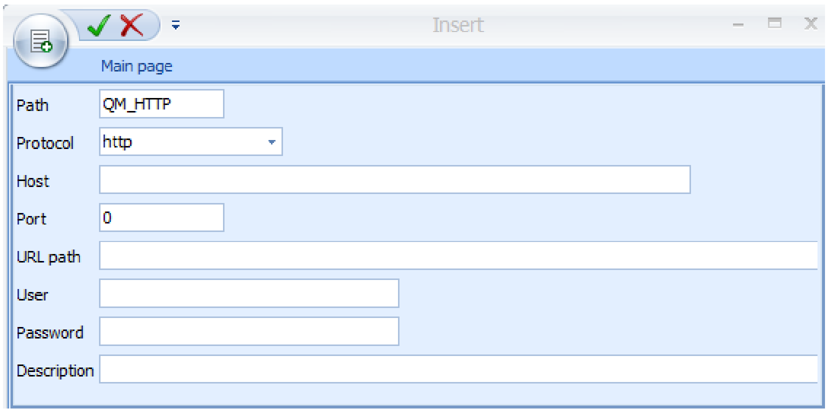
Figure: Document display window called in the *"Documents"* index tab

☞ Note	**Document type/ links** The linked program (AIP configuration) opens the document.

The display of documents in the CAQ is similar to the displays in MF (manufacturing).

The following rules apply:

- Documents of type `URL` are processed like order documents that were defined with the `http` path scheme.



The URL itself is taken directly from the document entry.

- For `Text` type document entries, the contents of the text fields 1 to 10 are written in a temporary text file. In the process, the masking for the line break is converted back into line breaks.
  This text file is then displayed "internally" (display switches between "open document" and "close document")

- `File` type documents are either opened directly or are requested from the HYDRA server and then opened. The following rules apply for the different methods:

  - <u>Paths below the CAQ documents directory</u>
    If a file name does not start with a dot, a slash or a backslash and if there is no colon in the second position (`D:\`) then the CAQ document path will automatically be placed in front of the file name.
    By default, this path is configured with `./caq/dokus/`.
    However, you can override this path at any time if you change an AIP INI option globally or for a specific terminal.
    This path can then be either an absolute or a relative path. In order to access the file, proceed as described below.

  - <u>Relative path details</u>
    If a file name starts with `./` or `.\` then the system assumes that it is a HYDRA subdirectory.
    In this case, the following part can only contain a file name (`./ReadMe.txt`) or a combination of subdirectories and the file name (`./cad_files/Artikel_0815.dxf`).

  - For document entries of this type, the system requests the file from the server and loads the file into the spool directory. The file is then opened either internally or externally.

- <u>Absolute path details</u>
  It is an absolute path:
  - If a file name begins with a backslash
  `(\\server1\vol2\files\CAQ_2345.txt)` or
  - a slash `(//server1/vol2/files/CAQ_2345.txt)` or
  - if the file name contains a colon in the second position
  `(M:\files\CAQ_2345.txt).`
  For document entries of this type, the system opens the file via direct internal or external access.

1.8.3 Process overview - variable

The application only shows this index tab if the current characteristic is *variable* and at least *Control chart #1* is defined.

The "variable" process overview is set up to display

- *Control chart #1*,
- *if required Control chart #2*,
- if required a *Histogram*
- and the most important *statistical values*

If *Control chart #2* is not available, then the size of *Control chart #1* is increased. It fills the area intended for both control charts.

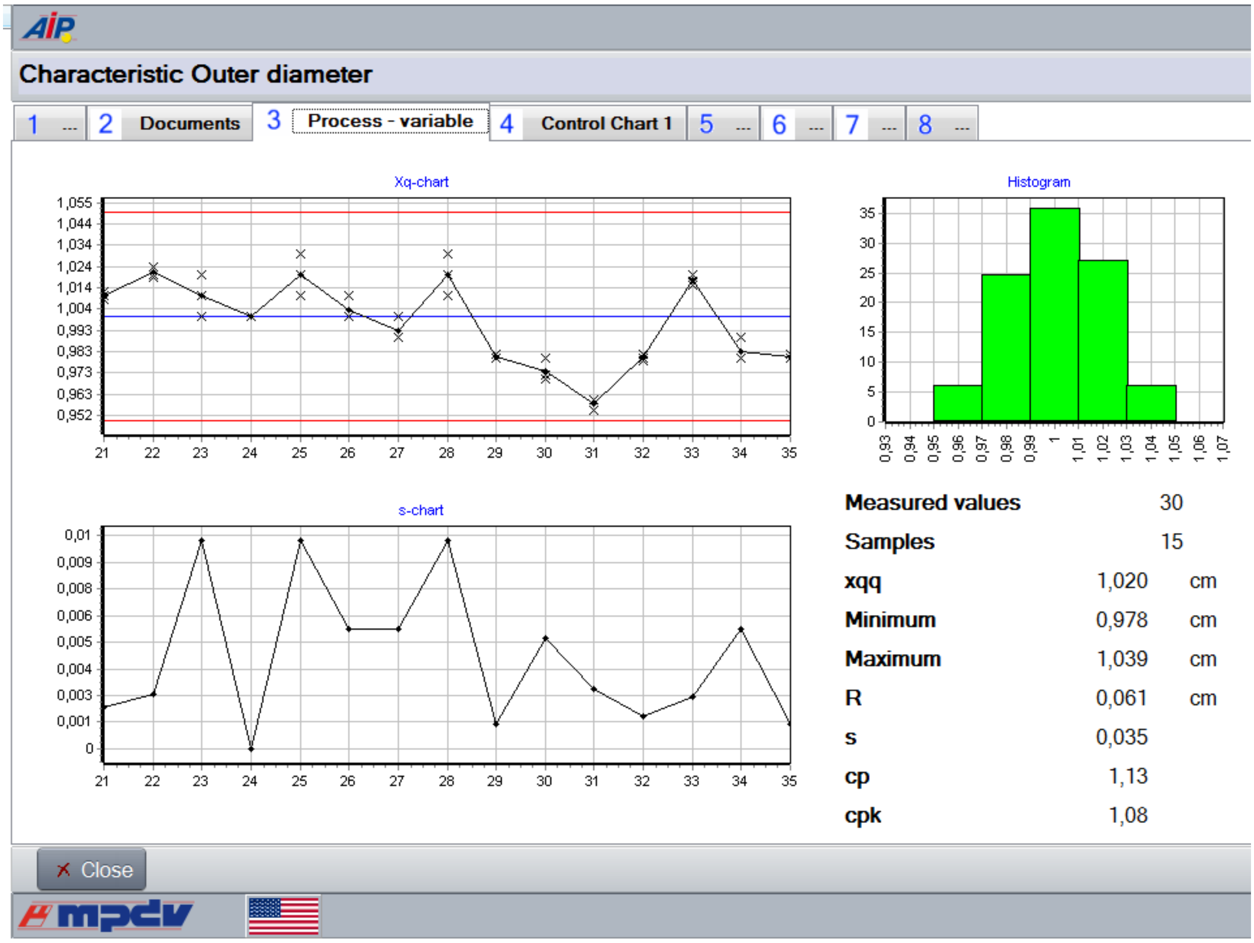


Figure: Index tab "Process - variable" in the dialog "information on characteristics" (with Control chart #1)

Note	**Displaying the histogram** The application only shows the histogram if data collection is based on *single values*.

	Exiting *information on characteristics*

1.8.4 Process overview - attributive

The application only shows this index tab if the current characteristic is *attributive* and at least *Control chart #1* is defined.

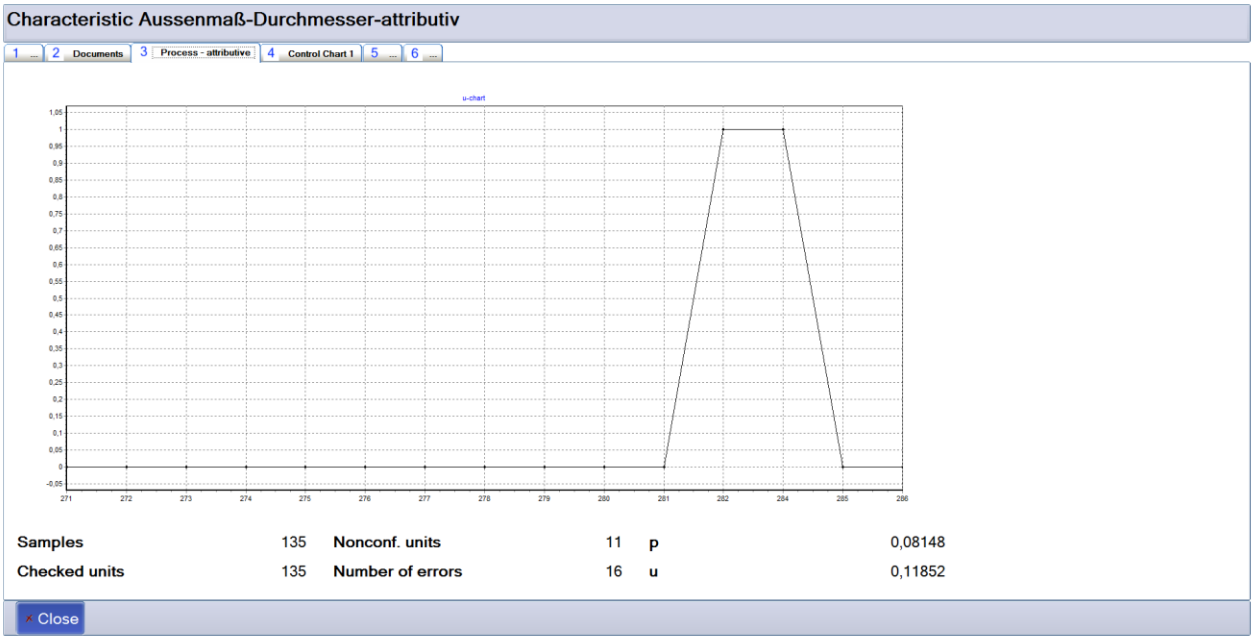


Figure: Index tab "Process - attributive" in the dialog "information on characteristics" (with Control chart #1)

	Exiting *information on characteristics*

1.8.5 Control chart #1

This index tab shows the *Control chart #1* defined for the characteristic and the limit values.

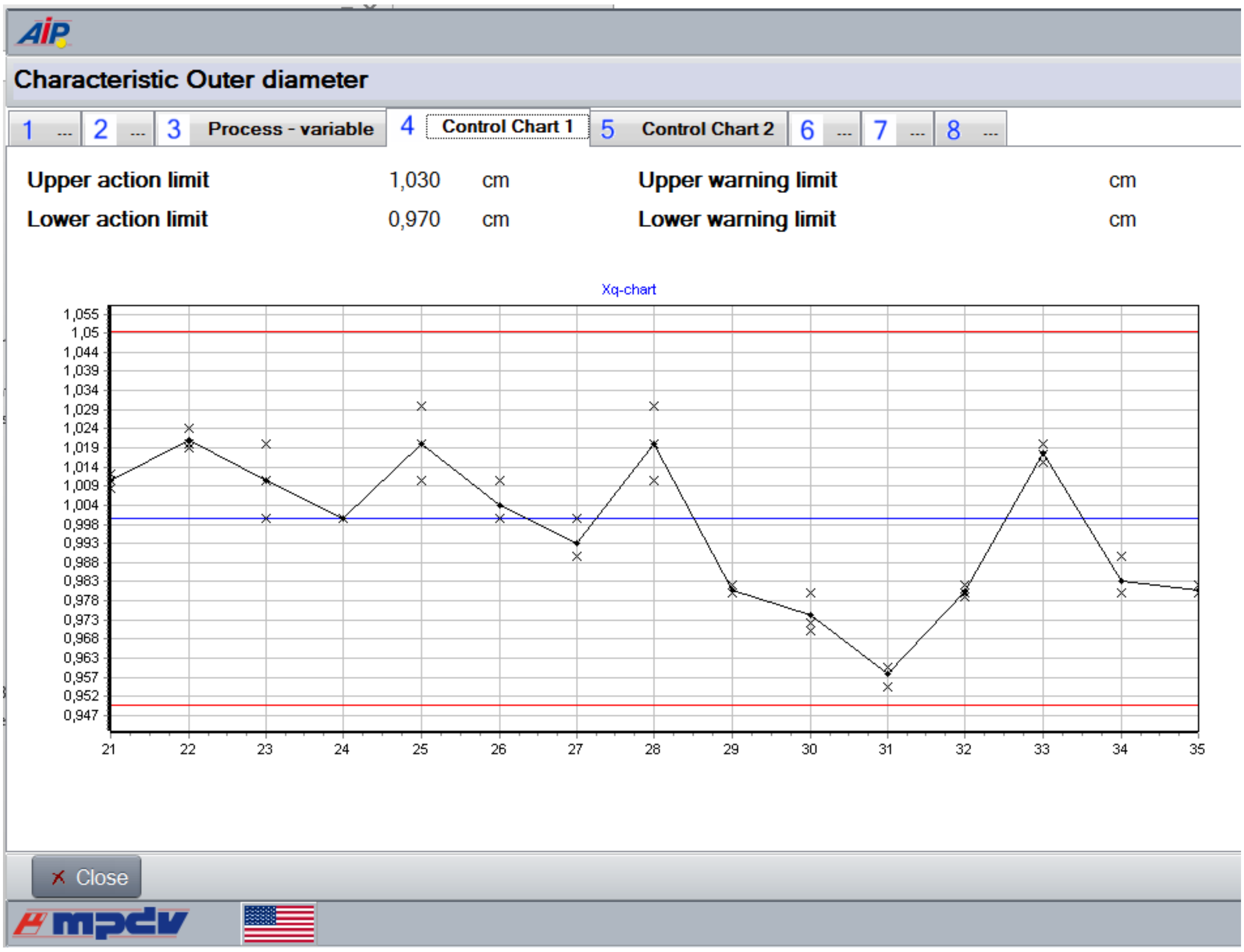


Figure: Index tab "Control chart #1" in information on characteristics

☞ **Note**	**Display of Control chart #1,** If no control chart #1 is defined for the characteristic, this index tab is hidden.

 	Exiting *information on characteristics*

1.8.6 Control chart #2

What is presented here is similar to what is shown in the index tab for control chart #1. However, here the default values and the progression of *Control chart #2* are visualized.

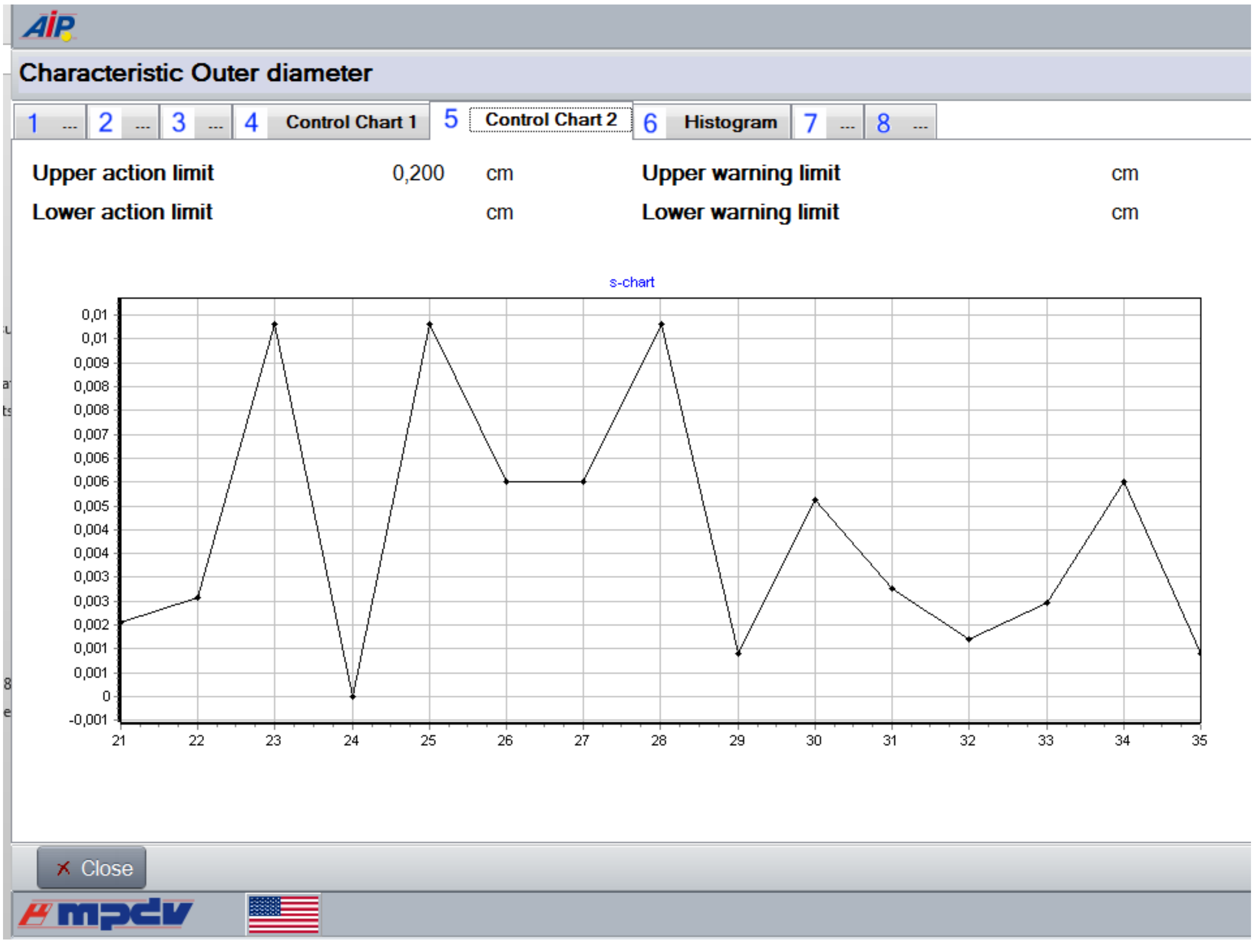


Figure: Index tab "Control chart #2" in information on characteristics

Note	**Display of Control chart #2,** If no control chart #2 is defined for the characteristic, this index tab is hidden.

	Exiting *information on characteristics*

1.8.7 Histogram

This index tab is hidden if the characteristic is not a *variable* characteristic, or if no input type with single-part inspection has been assigned to the characteristic.

☞
Note

Displaying the histogram

This index tab is hidden:

- if the characteristic is not a *variable* characteristic, or
- if no input type with single piece inspection was assigned to the characteristic.

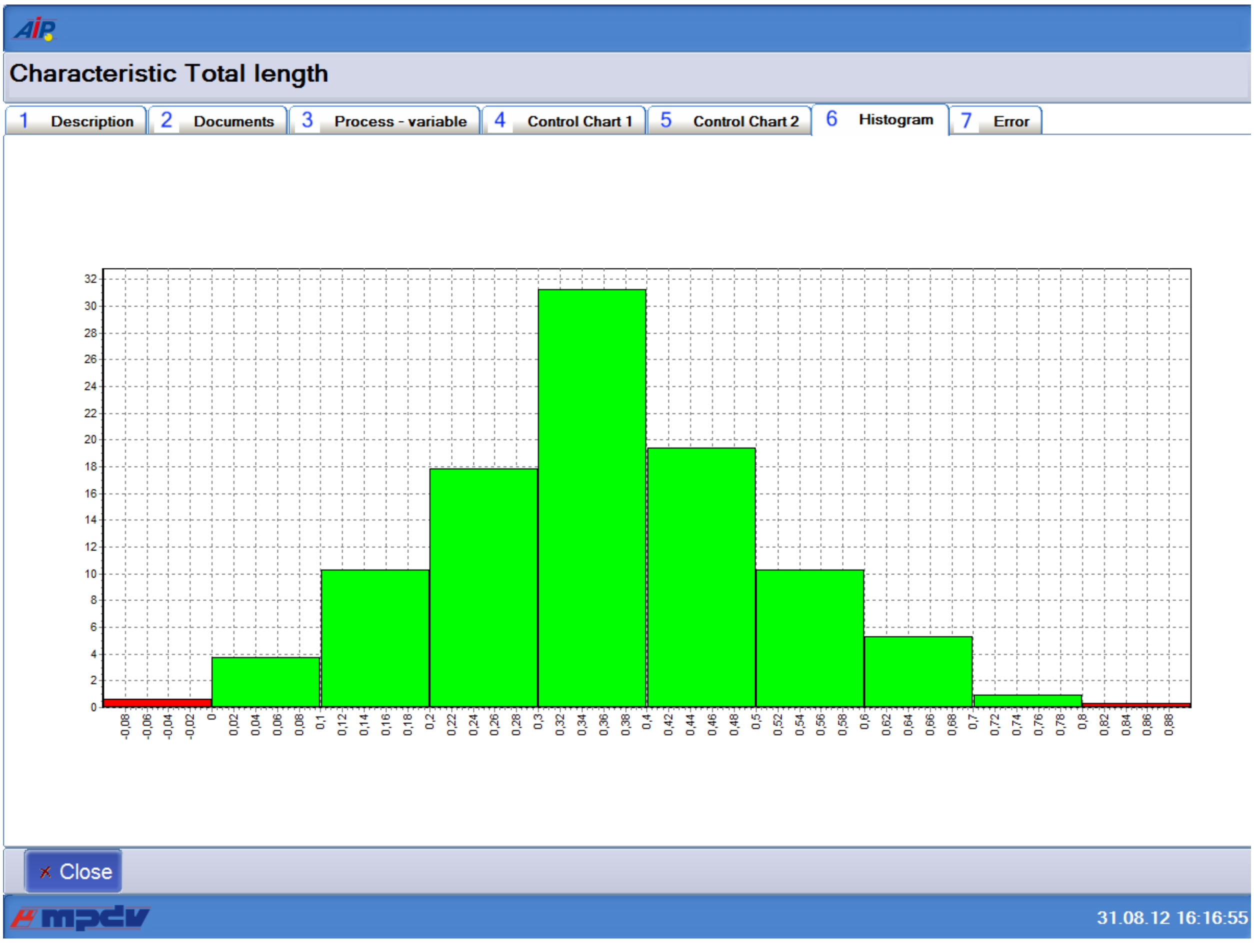


Figure: Index tab "Histogram" in information on characteristics

	Exiting *information on characteristics*

1.8.8 Failure history

This tab is only configured in the standard workflow definition of the characteristic information.

This index tab shows all entries of failure types assigned to the current characteristic or to global structures (inspection step / inspection requirement).
For inspection steps relevant to inspection points, this tab shows all entries of failure types that are assigned to the current inspection point (without characteristic assignment).

1.8.8.1 General failure history

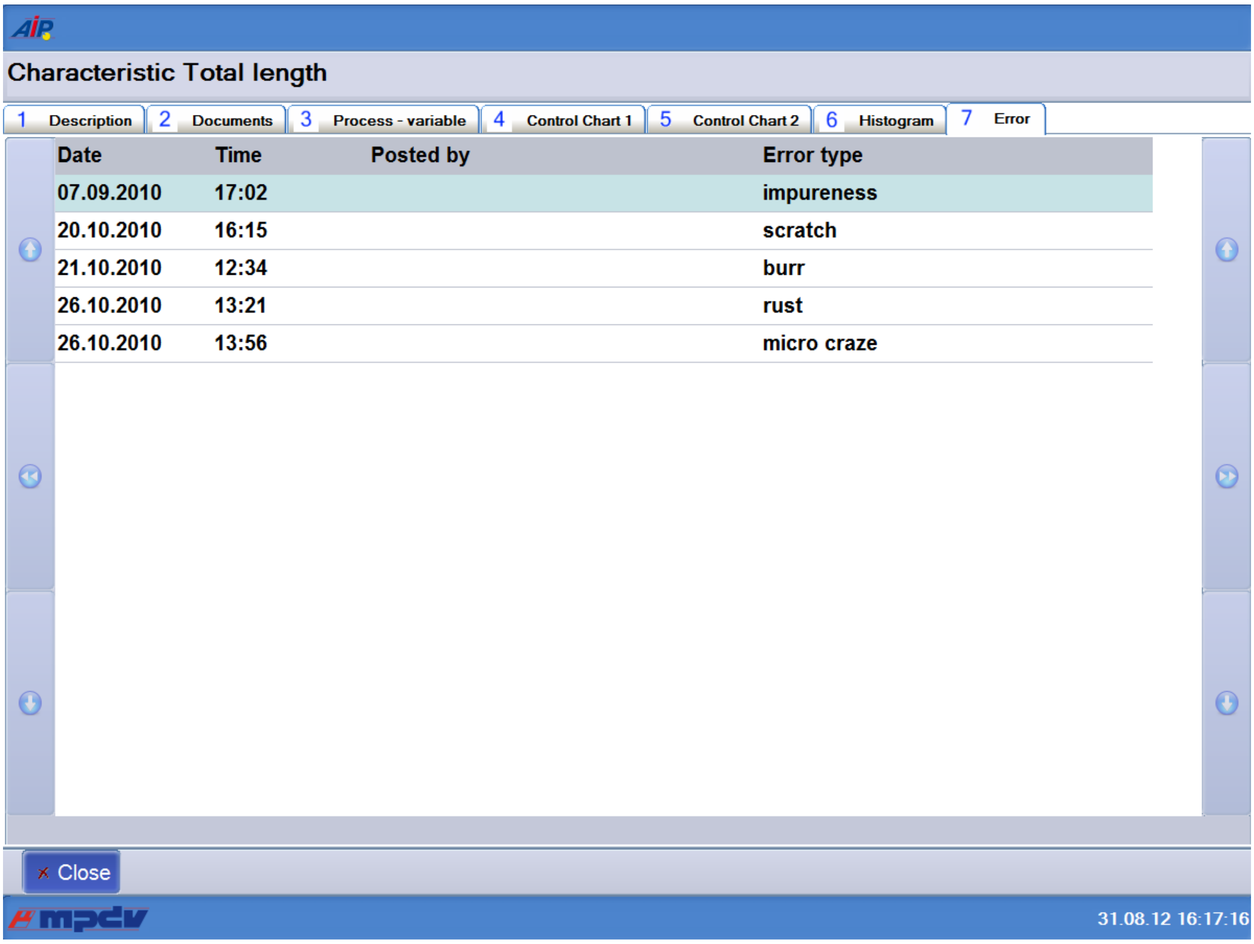


Figure: Index tab "failure history" in information on characteristics

	Exiting *information on characteristics*

1.8.8.2 Failure history for the visual assignment of failures (as of CAQ 8.2)

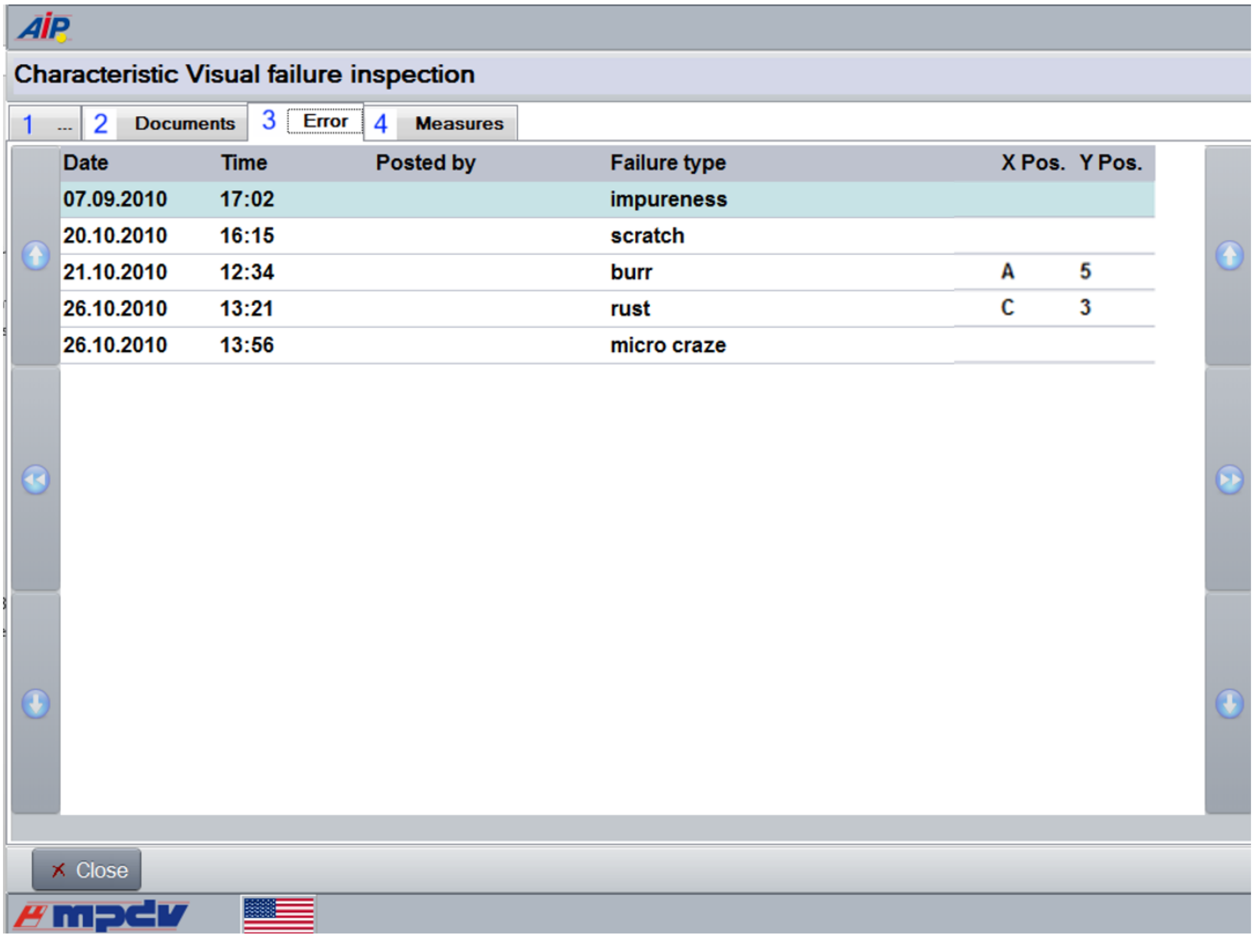


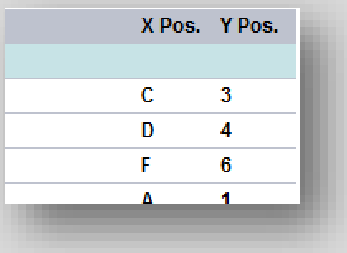
Figure: Index tab "Failure history, visual assignment of failures" in the information on characteristics dialog

Note

Extended display of the failure history

The failure history shows the following additional columns if the input type *"visual defects recording [JIT/JIS]"* is assigned to the selected characteristic.

- X pos.
- Y pos.



X Pos.	Y Pos.
C	3
D	4
F	6
A	1

→ see section Visual defects recording [JIT/JIS] as of MW 3.0

	Exiting *information on characteristics*

1.8.9 History of measures

This tab is only configured in the standard workflow definition of the characteristic information.

This tab shows all measures assigned to the current characteristic or global structures (inspection step/inspection requirement).
This tab also shows all measures belonging to the current inspection point (without characteristic assignment) for inspection steps relevant to inspection points.

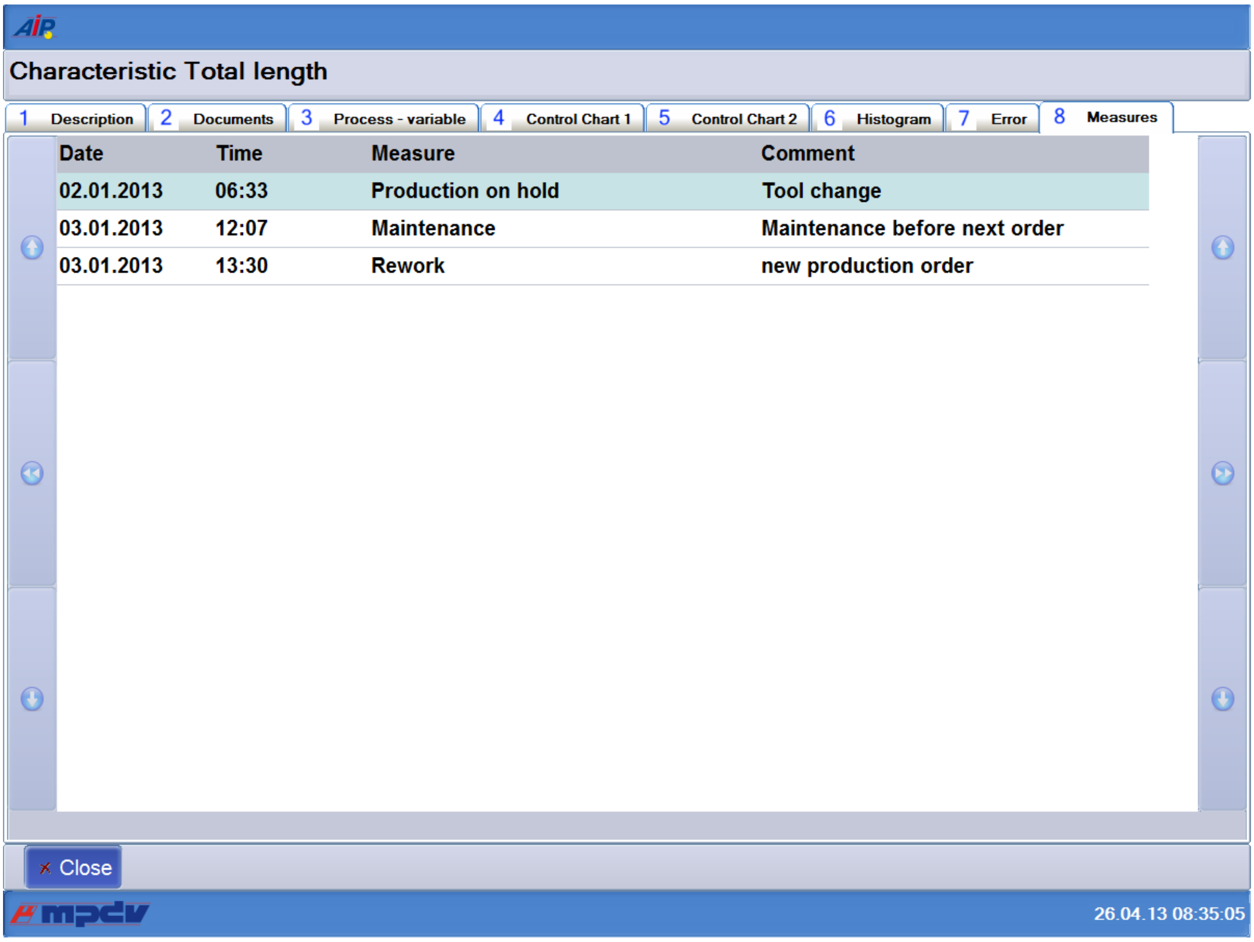


Figure: Index tab "History of measures" in the characteristics information dialog

	Exiting *information on characteristics*

1.9 MDI drivers

You can install and manage MDI drivers via the *Inst32* start menu.

1.9.1 Backup of MDI drivers

Select the following entry in the main menu:

[E] Tools

Go to the first submenu and select the following entry:

[5] more Tools

In order to switch to the next submenu, as a final step select:

[1] Backup

There will first be a query asking whether you would like to run a backup.

For the backup, the server uses the file **<hydradir>**\ctnet\win\ctaipbackup.txt
or a terminal-specific file **<hydradir>**\ctnet\win\ctaipbackup2xxx.txt
(xxx is the terminal number).

First, the system attempts to load a terminal-specific file.
If no terminal-specific file exists, the system will then attempt to load the file ctaipbackup.txt.
This file contains all of the files or registry entries that need to be backed up.

Example:

```
\ctaip\*.INI
\ctaip\*.cfg
\ctaip\cfg\*.*
HKEY_LOCAL_MACHINE\SOFTWARE\MPDV\
```

The configuration files for an MDI driver are usually located in its installation directory and typically have INI as the file extension. Refer to the driver's documentation to learn which configuration files are relevant for which drivers. Below is an example of how to back up settings of two separately installed MDI drivers:

```
\mdi\Steinwald\*.INI
\mdi\Messwertdatei\*.INI
```

The terminal generates a Zip file and stores it in the server.
The file is located in the server under:
The backup Zip file is given the name: ctaipbackup2xxx.zip ->xxx = terminal number (terminal-specific for Hydra user 2xxx)
This backup file is then stored in the server under

<hydradir>\custom\backup\ctaip\ctaipbackup2xxx.zip .

Restore MDI drivers

Select the following entry in the main menu:

[E] Tools

Go to the first submenu and select the following entry:

[5] more Tools

In order to switch to the next submenu, as a final step select:

[2] Restore

There will first be a query asking whether you would like to run a restore.

"Restore" tries to load a backup file located in the server and then automatically restores all the backed up files and any backed up registry entries.
A backup file is stored on the server in the directory:

For a single system installation:
<hydradir>\custom\backup\ctaip\ctaipbackup2xxx.zip

For a multi-system installation:
<hydradir>\<*system*>\custom\backup\ctaip\ctaipbackup2xxx.zip

as already described under backup.

Installing 3rd party MDI drivers

Select the following entry in the main menu:

[E] Tools

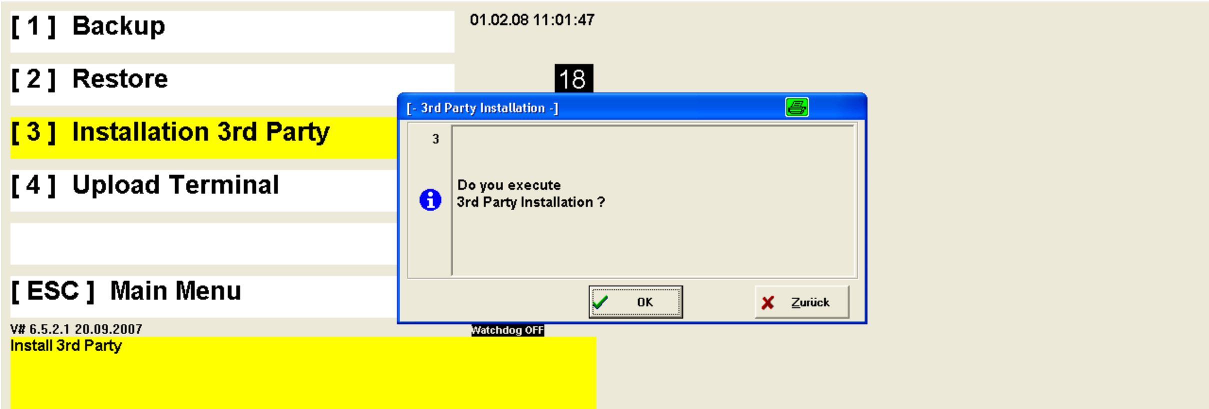
Go to the first submenu and select the following entry:

[5] more Tools

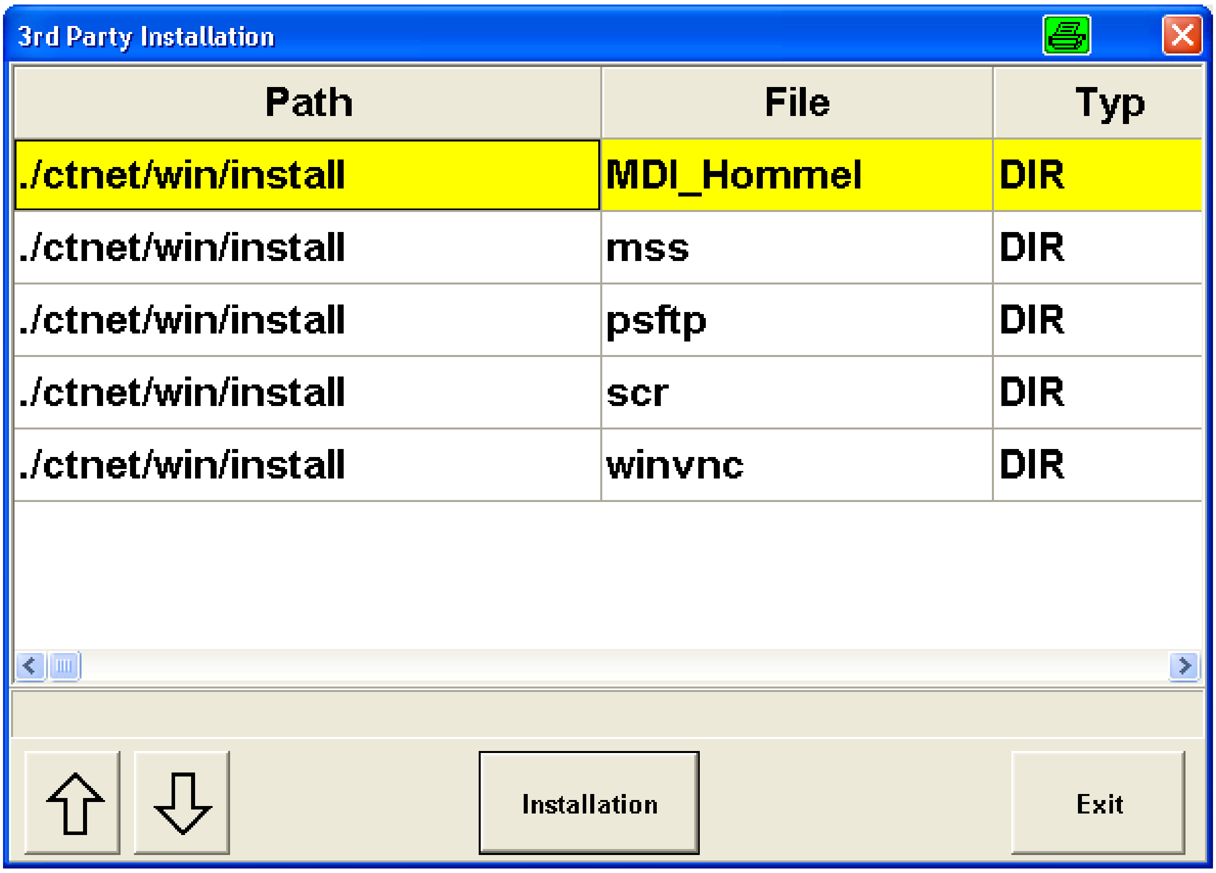
In order to switch to the next submenu, as a final step select:

[3] Installation 3rd Party

There will first be a query asking whether you would like to perform a 3rd party installation.



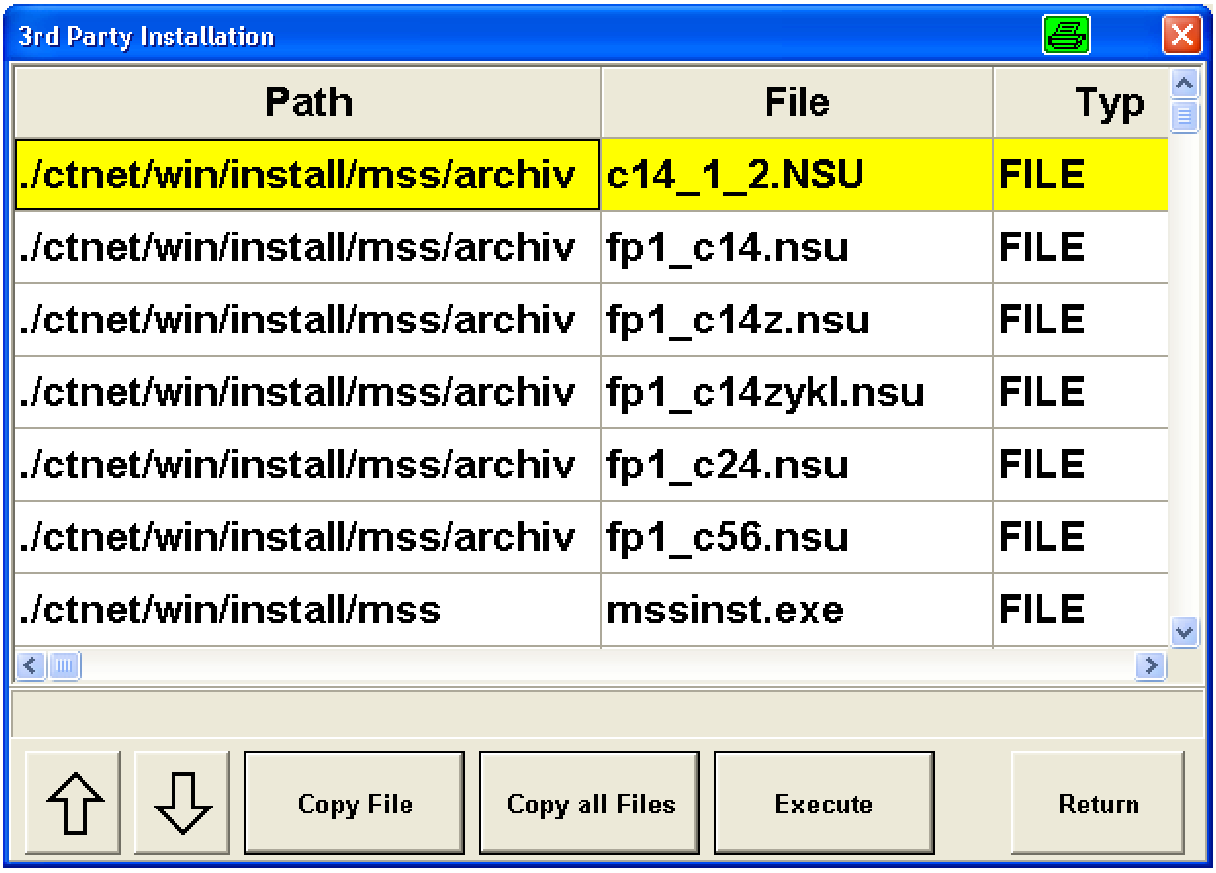
After confirming the query by clicking on **OK**, the following list will appear showing the directories found in **<hydradir>\ctnet\win\install**.



Installation button:

Click the **Installation** button to display a list indicating all of the directories starting with directory **<hydradir>\ctnet\win\install**.

The directories found will be offered for selection in a dialog. Once you have confirmed a directory, the system downloads this directory and shows its contents.



Check whether the directory of the MDI driver you would like to install is included in a directory. If not, load the MDI installation in the appropriate subdirectory of **<hydradir>\ctnet\win\install**.

Exit button:
Click *Exit* to cancel the process and return to the main menu.

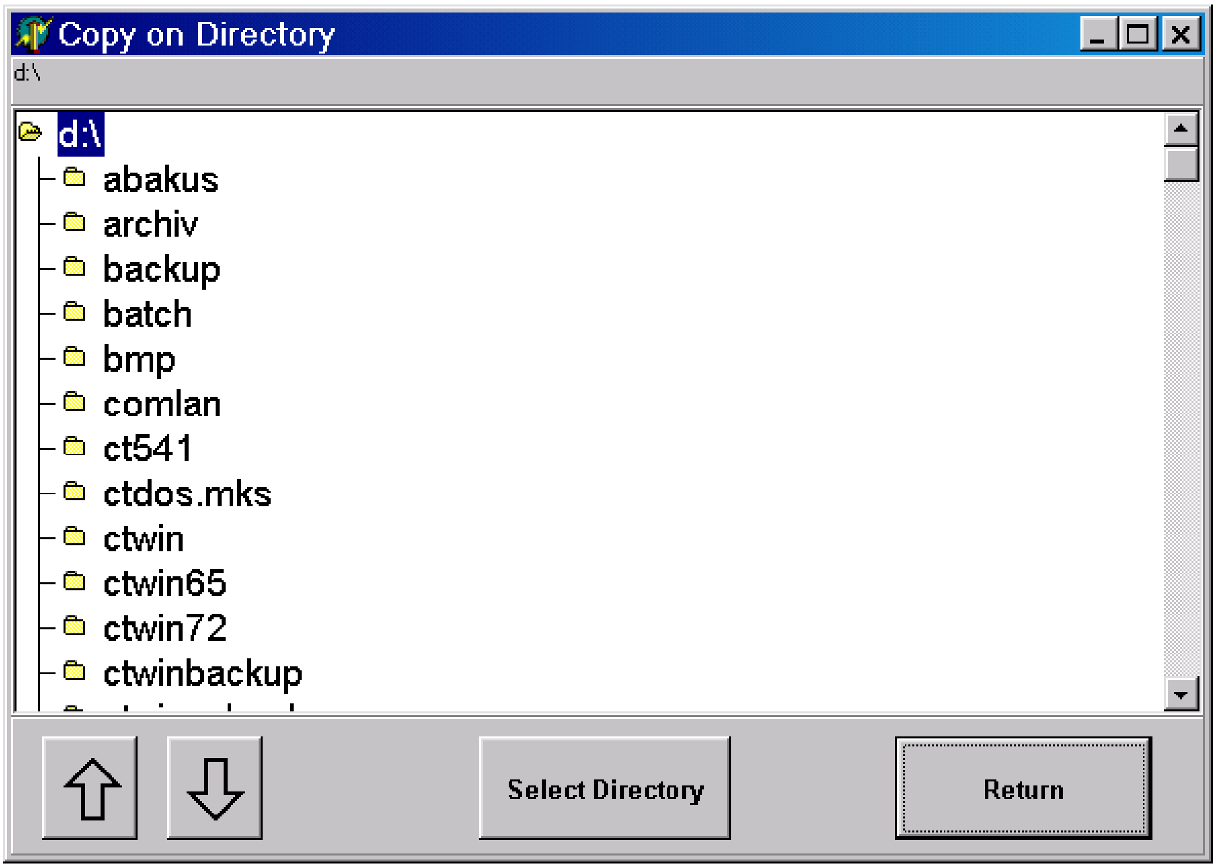
Therefore, you can add further directories to the directory **<hydradir>\ctnet\win\install.**

Content of a previously selected directory

Button "Copy File" / "Copy all Files"
After clicking on one of the two buttons "**Copy File**" or "**Copy all Files**", a selection window will appear to select a directory into which you can now copy the selected file or all files being displayed. Select the file first, if you would like to copy a single file from the list.

Selection window to select a directory

Return button:
Press the Return key to return to the selection of directories



Execute button:

Click this button to start the installation process of the MDI driver. The execution program defined in Windows is used to display or execute the selected file.

Example: Installing the Steinwald MDI driver

<u>Activity in preparation (once)</u>

Copy the Steinwald installation files into a server sub-directory (e.g. "mdi_steinwald") **<hydradir>\ctnet\win\install**.

<u>Installation on the terminal (must be installed for each terminal)</u>

0. Start the **start menu Inst32**

1. In the main menu, select **[E] Tools**

2. In the first submenu, select **[5] more Tools**

3. In the next submenu, select **[3] Installation 3rd Party**

4. Confirm the security prompt that appears next with **OK**

→ You will see a list with the sub-directories located in the **<hydradir>\ctnet\win\install** directory on the server.

5. Now, select the "mdi_steinwald" folder and click the **Installation** button to continue the process (optionally, you can click the **Exit** button to return to the previous submenu).

→ You will see an overview showing all of the files found in the "mdi_steinwald" folder.

6. In a next step, store all of the files from the "mdi_steinwald" folder locally by clicking on the **Copy all Files** button.

7. Now, select the desired target location where you would like to copy the files.

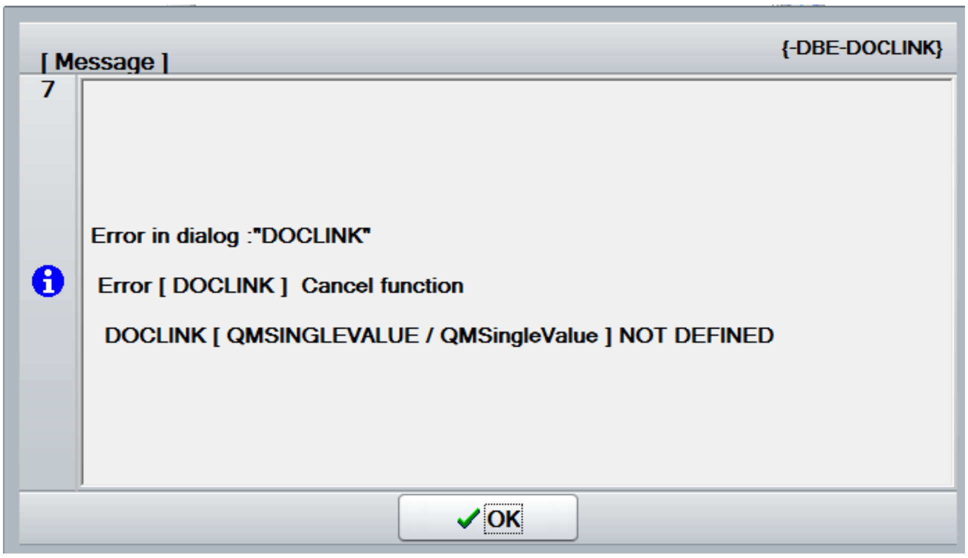
8. To start copying the files, press the **Select Directory** button (optionally, you can click on the *Return* button to return to the previous dialog without copying the files). When copying is completed, the dialog closes.

9. Select the setup program in the overview to install the driver and confirm by clicking the **Execute** button.

10. Execute the installation process of the Steinwald MDI driver.

1.10 Document management (as of CAQ 8.2)

The terminal shows the following message if *document management* is started and **inspection results recording has not yet been saved.**



For further information on the *document management,* refer to the respective manual.

1.10.1 Configuration / inspection point

You can manage documents

- in relation to an inspection point.

Configuration

Terminal configuration file `caq_dc_t.ini / caq_dc_t.*`

- Layout definitions for buttons of the CAQ inspection list

```
[CAQ_DC_T-PPKT-Page3]
1=$DOC-LINK$DOCLINK.InspectionPoint,L,Dokumente Pruefpunkt
```

For space reasons, we recommend to place the buttons for the document management on a new, empty page (here page 3).

Figure: Layout for buttons added to the inspection list/document management in relation to an inspection point

1.10.2 Configuration / attributive data collection

As part of the attributive data collection, you can manage documents for the

- inspection step characteristic
- inspection point characteristic
- inspection result

Configuration

Terminal configuration file `caq_dc_t.ini / caq_dc_t.*`

- Layout definitions for buttons of the CAQ inspection list

```
[CAQ_DC_T-STICHPR-Page3]
1=$DOC-LINK$DOCLINK.InspectionCharacteristic,L,Dok. Merk.-Pruefschritt
2=$DOC-LINK$DOCLINK.InspectionPointCharacteristic,L,Dok. Merk.-Pruefpunkt
3=$DOC-LINK$DOCLINK.QMSingleValue,L,Dok. Pruefergebnis
```

For space reasons, we recommend to place the buttons for the document management on a new, empty page (here page 3).

Figure: Layout for buttons added to the inspection list/document management in relation to an attributive characteristic

1.10.3 Configuration / variable data collection

In relation to a variable characteristic you can manage documents for the

- inspection step characteristic
- inspection point characteristic

In relation to inspection results recording you can manage documents for

- the inspection result (measured values, attributive assessments, inspection chart)

Configuration

Terminal configuration file `caq_dc_t.ini / caq_dc_t.*`

- Layout definitions for buttons of the CAQ inspection list

```
[CAQ_DC_T-ESTCK-Page3]
1=$DOC-LINK$DOCLINK.InspectionCharacteristic,L,Dok. Merk.-Pruefschritt
2=$DOC-LINK$DOCLINK.InspectionPointCharacteristic,L,Dok. Merk.-Pruefpunkt
3=$DOC-LINK$DOCLINK.QMSingleValue,L,Dok. Pruefergebnis
```

For space reasons, we recommend to place the buttons for the document management on a new, empty page (here page 3).

Figure: Layout for buttons added to the inspection list/document management in relation to a variable characteristic

Figure: Layout for buttons added to the inspection list/document management in relation to inspection results recording for variable characteristics

1.11 Transferring measured values for all characteristics (as of CAQ 8.2)

Transferring measured values irrespective of the characteristic is only available for ***variable characteristics in relation to an inspection point.***
These characteristics must not be in the status "skip lot". The characteristics must have the status "can be checked", "checked" or "result".
MDI processing does not integrate characteristics with a calculation formula.

This function *transfers measured values for all characteristics.*

In contrast to the (individual) collection of inspection results on the terminal, this function enables processing of

- *a multitude of measured values relating to an inspection point*
- *for several characteristics*
- *triggered by a single user action*

Organizational requirements

The user must ensure that the number of measured values available for the data transfer does not exceed the sample size of the characteristic.

Performance

This function does not reduce the time that is required to save measured values. This function "only" automates the processing/save processes.

Consequently, the time required for processing/saving depends on the number of measured values.

1.11.1 Configuration in HYDRA

In relation to an *inspection point,* you can transfer measured values for all characteristics.

Configuration

Terminal configuration file `caq_dc_t.ini / caq_dc_t.*`

Please add/change the following in the terminal configuration file `caq_dc_t.ini / caq_dc_t.*` in order to transfer measured values for all characteristics:

- Layout definitions for buttons of the CAQ inspection list

```
[CAQ_DC_T-PPKT-Page2]
1=DQC_TRANSFER_DATA,L, accept measurement data
2=DQC_RELOAD,R, update display
```

➔ The modifications become effective, once you have restarted the terminal.

Figure: Layout for buttons added to the inspection list/transferring measured values for all characteristics in relation to an inspection point

1.11.2 Function description / operating instructions

If you click the button *"Accept measurement data",* you start the *transfer of measured values for all characteristics* in the *inspection list.*

Then the HYDRA server immediately processes the data.

Data is always transferred in relation to the *inspection point currently selected in the inspection list.*

☞ **Note**	**Behavior in offline status** Processing does not take place in the offline status, as an online connection is required between the terminal and the HYDRA server.

The terminal is still available to the user, while the HYDRA server is processing the measured values. You can continue collecting (individual) inspection results, if necessary.

☞ **Note**	**Updating of data** You can only manually update inspection results on the terminal. No message appears when the server finishes processing.

☞
Note

The following scenarios are possible because the processing is independent of other inspection activities:

Scenario 1:

- The terminal user triggers the transfer of measured value for all characteristics on the AIP and continues with another task. But measurement recording still remains open.
- The HYDRA server processes the measured values. But the program does not update the displayed data.
- After the lunch break, the user then sees an outstanding inspection point on the terminal and wants to enter results.

  After *transferring measured values for all characteristics,* the terminal does not automatically display the recorded values. The user must *update the display manually.* If no update is performed, a message informs that the value is already available.

➔ **Solution: Click the button *Update display***

Scenario 2:

- The user triggers the measured value transfer for all characteristics on the AIP and manually enters further inspection data.
- The inspection will be completed and/or is supposed to be completed.

➔ However, it is possible that not all measured values have been processed entirely when the inspection point is completed. The inspection point cannot be completed if the minimum sample size, e.g. for mandatory inspections, has not been reached.

Before measured values for all characteristics are accepted, the system checks if: the *inspector has been identified* when starting the *CAQ inspection results recording.* And if a *staff badge number* has been defined for *CAQ inspection results recording.*

If this is not the case, the application shows the dialog for the *inspector identification*.

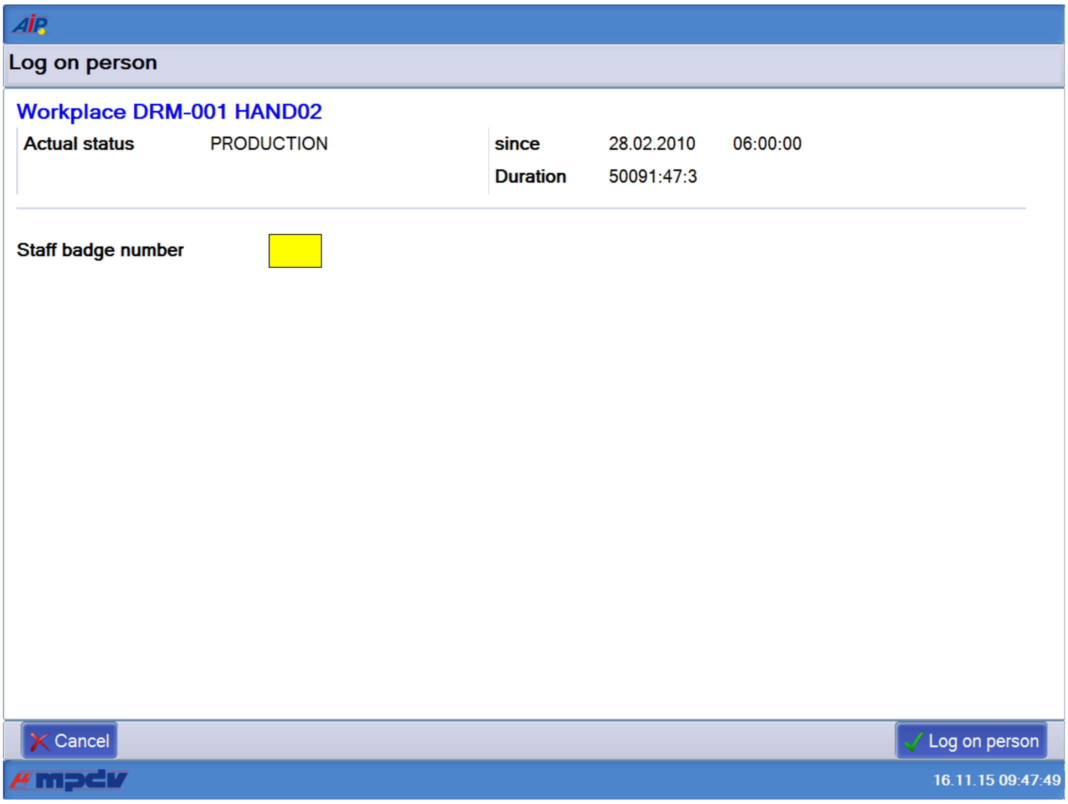


Figure: Dialog to enter the staff badge number

Click "OK" to accept the *measured values for all characteristics.*

Click the button "cancel" to exit the dialog without processing the data.

☞
Note

Technical requirements

Ensure the following to establish communication with all MDI drivers:
- the HYDRA server must be able to communicate via TCP/IP and the corresponding communication port with the computer where the MDI driver is running.

☞
Note

Logging

In case of errors/issues, the HYDRA server generates log files in the log directory (*<system>*\prot).
File names of log files have the following structure: hy_cmdilrv_*AUFTRAG_CAPTURE*_*.csv. Store log files in the CSV format and view the files e.g. with Excel.

☞
Note

Processing of measured values by MDI driver

The system only processes confirmed MDI measured values (`CONFIRMED=1`). The system always deletes unconfirmed MDI measured values (`CONFIRMED=0`) from the MDI buffer without posting the data in HYDRA.

The system checks if the MDI measured values respect the validation limits of the characteristic. If a measured value violates these limits, the measured value is deleted (if MDI is configured accordingly). If you do not want to delete the MDI measured value, processing of the current characteristic is stopped.

The system stores the following MDI parameters (if available):

`SERIAL`	Unique identifier of the MDI measured value
`MVALUE`	Actual measured value This is a mandatory parameter.
`MDATE`	Date of data collection
`MTIME`	Time of data collection
`MFROM`	Inspector name (instead of badge number)
`MTEXT`	Comment
`NEST`	Cavity number The application only saves this parameter, if you want to record the characteristic relating to cavities on the "sample" level. In this case, the cavity number is a mandatory parameter.

The following AIP data is also stored:

- The inspector's badge number
- Machine/workplace number

☞ **Note**

Requesting measured values from MDI driver

The system requests the measured values from the MDI drivers. The system only returns and processes the measured values matching the following filter criteria.

`ANR`	Order number of the inspection requirement
`AGNR`	Operation number of the inspection step (primary) or the inspection requirement (secondary).
`ATK`	Article number of the inspection requirement
`CNR`	ERP batch of the inspection requirement
`MNR`	Workplace where inspections are currently performed.
`PPKT:TLOS`	Partial batch of the inspection point.
`PPKT:CNR`	ERP batch of the inspection point.
`PPKT:EQUIP`	Tool of the inspection point (MOC inspection point list: field 1)
`PPKT:PROBE`	Sample of the inspection point (MOC inspection point list: field 3)
`PPKT:USERC1`	User field C1 of the inspection point (MOC inspection point list: field 4)
`PPKT:USERC2`	User field C2 of the inspection point (MOC inspection point list: field 5)
`PPKT:USERN1`	User field N1 of the inspection point (MOC inspection point list: field 6)
`PPKT:USERN2`	User field N2 of the inspection point (MOC inspection point list: field 7)

1.12 Saving measured values using the ENTER key (as of CAQ 8.2)

In some dialogs you can confirm inspection results directly by pressing the ENTER key of the (hardware) keyboard.

But you can also click the "Done" button.

1.12.1 Configuration in HYDRA

Configuration	**Terminal configuration file** `hytnrcfg.ini`
	Change the terminal configuration file `hytnrcfg.ini` as described below in order to use the Enter key to confirm the data input (simulating the "done" button):
	Insert the following rows
	`[DYNAMIC-DIALOG->Options 0]`
	`USE_ENTER_BUTTON=1`
	
	Default → 0
	➔ Restart the respective terminals to enable the modification.

The following list provides an overview of the dynamic dialogs where you can use the ENTER key:

- `QEE_MW_ME_ES_PP_SI`
- `QEE_MM_BE_ST_PP_SI`
- `QEE_MM_BE_ST_PP_FS`
- `QEE_INSPPOINT`
- `QEE_INSPPOINT_DETAIL`
- `QEE_MM_PR_PP_SI`
- `QEE_MASS_CLASSIC`
- `QEE_ERR_CLASSIC`
- `Q_P_AN`

1.12.2 Function description / operating instructions

Pressing the ENTER key triggers the "Done" button.

☞
Note

The following applies ...

The ENTER key is triggered irrespective of

- the currently focused field of the *data input panel*
- the currently focused area of the GUI (left: *inspection list*, right: *data input panel*).

☞
Note

Exceptions

The ENTER key is not triggered if the focus is on

- one of the buttons of the *inspection list*
- another button of the *data input area*
- one of the *selection list buttons* of the data input area

has the focus

1.13 Integration of CAQ-MPL/TRT (as of CAQ 8.2)

☞
Note

Transferring batch information to the inspection point

When an inspection point is generated, the system enters MPL batch information in the "ERP batch" field of the inspection point (`PPKT:CNR`, 50 characters).

☞
Note

Properties of MPL batch fields

In general, you can enter up to **40 characters** in the MPL batch fields *Alternative batch number 5 to 20.*

But the AIP can only display up to 20 characters. You must therefore make sure that the used MPL batch fields *alternative batch number 5 to 20* **only** include **20 characters.**

☞
Note

Further restrictions of the CAQ-MPL integration

It is possible that the system assigns the "wrong batch" or that the system cannot uniquely identify the machine if you have manually generated an inspection point.

1.14 Preceding list of inspection points (as of CAQ 8.2)

mpdv-aip.zip	*>=2015-06-22*
aip-qm.dlg	*>=2015-06-22*

This function is available as of CAQ 8.2 but not enabled by default.

The following configuration description explains how to enable the function.

If customizations are in use, this function can be restricted and/or disabled.

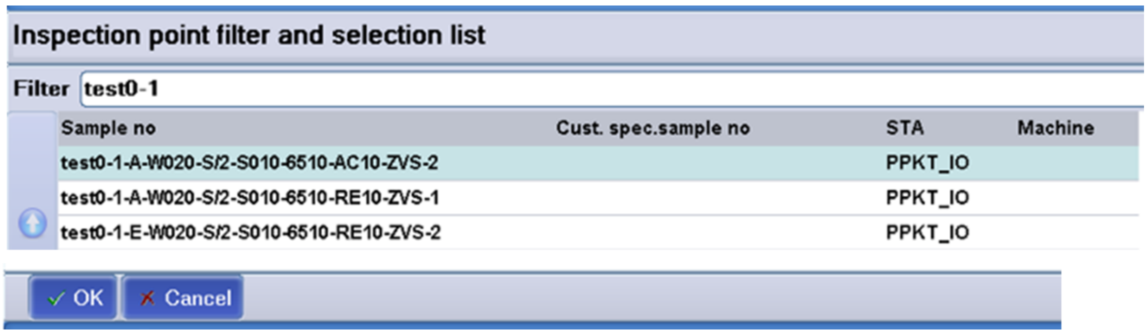


Figure: Preceding inspection point list

Go to the inspection list in order to generate inspection points manually. Once generated, exit the inspection list, select the new inspection point and reopen the inspection list.

1.14.1 Configuration

Once you have enabled the preceding inspection point list, the "update" button of the inspection list does no longer work. Close and reopen the inspection list in order to update its data.

You can enable the preceding inspection point list for specific terminals only if you define terminal-specific configurations (CAQ --> Option 99 for terminal 99).

Configuration

Terminal configuration file `hytnrcfg.ini`

In order to use the "preceding inspection point list" on the terminal, make the following changes to the terminal configuration file `hytnrcfg.ini`.

Insert the following rows

```
[CAQ->Optionen 0]

LOAD_MEASUREMENTS_ON_DEMAND=ON
INTERPOSE_FUNCTION=QEE_FILTER_INSPPOINT
RECALL_ON_EXIT_INSP_LIST=ON
REQUEST_RELOAD_ON_EXIT_INSP_LIST=MNR,ANR
```

➔ Restart the respective terminals to enable the modification.

Note:

You must set the option "`LOAD_MEASUREMENTS_ON_DEMAND`" to "ON" if you use the preceding inspection point list.

If the option "`INTERPOSE_FUNCTION`" does not include a parameter or if the entry does not exist, the inspection point list cannot be reopened automatically after you exited the inspection. Enter the dialog that is called as parameter.

If the option "`RECALL_ON_EXIT_INSP_LIST`" is set to "OFF" or if the entry does not exist, the inspection point list cannot be reopened automatically after you exited the inspection.

If you configured the option "`REQUEST_RELOAD_ON_EXIT_INSP_LIST=`" with "`MNR,ANR`", the application updates the order and machine list, once you have exited the preceding inspection point list. This also updates the inspection status. If you only configure the parameter "ANR", the application only updates the order list. If this option does not exist, the order and machine list will not be updated when you exit the preceding inspection point list. Configure the order and machine list display if you want to display the order and machine list instead of tiles.

Configuration

Terminal configuration file `hytnrcfg.ini`

If you exit the inspection list and directly go to the machine and order list, these lists and the inspection status are not updated. Make the following entry to configure this behavior:

[DLL_DLG 0]

DISABLE=MNR,ANR,PNR,MSTAT,BPOS,AGRD,RES,LOKVLIST,ZLO,HZTYP,LICENSE,AART,PATHS,MAT,LPKZ,TPE,SKAL,QRD,PAUMNR,PPKTMNR

This entry ensures that the lists are not updated during the inspection. When the inspection list is closed, the specified lists are updated.

Important:

The configured lists are not updated until the inspection list is closed – whatever the consequences. For example, if the shift list is not loaded for a very long time, at some point in time the MDE processing runs out of shifts. In this case, the system cannot post a shift change that is due.

1.15 General notes

This section provides general information on the functionalities provided by the CAQ recording of inspection data recording with the AIP.

1.15.1 Field length in dynamic dialogs

Note the following if you use dynamic dialogs:

Note

Data origin: interfaces or MOC

In general, dynamic dialog data deriving from interfaces or the MOC can be displayed in an abbreviated form.

Subject to the origin of data, texts in dynamic dialogs can be displayed in an abbreviated form. **Note: After saving, these shortened contents are integrated irrevocably in the system.** The shortening and saving also applies for fields that cannot be edited, i.e. their content is only displayed.

☞
Note

Configuration options

If the text of fields (displayed or entered) must be abbreviated, you can still adjust the size of the single fields via configuration.

You can even move a field to another line to take up the full length of the dynamic dialog.

This applies for fields that are *saved* or *only displayed.*

Please note that input fields, which are hidden or removed, are ALSO saved

➔ Please contact your MPDV CAQ consultant.

☞
Note

Effects of configuration: field size

Changes to the field size can have considerable effects on the layout of a dynamic dialog.

☞
Configuration

Configuration options

If you change field sizes, you can adjust the aspect ratio between the “inspection list” and the “input panel” to harmonize the appearance of the user interface “inspection results recording”.

→Please contact your MPDV CAQ consultant.

List of (text) fields that are automatically saved by default:

Designation (name)	**Dynamic dialog**	**Field/ID**	**Field length: HYDRA DB**	**Field length: Dynamic dialog**
Inspection point	`QEE_INSPPOINT`	`CPANUMP.PPKT:USERC1` / *dynamic label from* *`PAU.PPKT:USERC1LAB`*	50	15

		CPANUMP.PPKT:USERC2 / *dynamic label from* *PAU.PPKT:USERC2LAB*	50	15
		CPANUMP.PPKT:EQUIP / *dynamic label from* PAU.PPKT:EQUIPLAB	20	15
		CPANUMP.PPKT:TPLATZ / *dynamic label from* PAU.PPKT:TPLATZLAB	20	15
		CPANUMP.PPKT:PROBE / *dynamic label from* PAU.PPKT:PROBELAB	20	15
Inspection point details	QEE_INSPPOINT_DETAIL	CPANUMP.PPKT:TLOS / *Partial batch*	50	15
		CPANUMP.PPKT:CNR / *ERP batch*	50	15
		CPANUMP.ENT:GRUPPE / Group	10	11
		CPANUMP.ENT:CODE / Code	10	5
Attributive data collection	QEE_MM_BE_ST_PP_SI QEE_MM_BE_ST_SI	CPAUMW.BEM / Comment	250	26
Variable collection	QEE_MW_ME_ES_PP_SI QEE_MW_ME_ES_ST_SI	CPAUMW.BEM / Comment	250	29
		CPAUMW.NEST / Cavity	50	10
Inspection chart	QEE_MM_BE_ST_PP_FS QEE_MM_BE_ST_FS	CPAUMW.BEM / Comment	250	26
Failure data	QEE_ERR_CLASSIC	CPAUERR.ERRNR / Number	50	15

Measures	`QEE_MASS_CLASSIC`	`CMASSN.MASNR` / Number	50	15
		`CMASSN.MASTEXT` / Measure	250	30
		`CMASSN.BEM` / Comment	250	30
Sampling	`QEE_MM_PR_PP_SI`	`PRBGRP` / Sampling	50	26

1.15.2 Processing and display of automatic failures

You can enable/disable the processing and display of automatic failures on the AIP.

→ See *Option 1214* in the document Configuration_QM_Options.docx

1.16 Asynchronous collection of measured values and failures (as of CAQ 8.1 and CAQ 8.2)

mpdv-aip.zip	*>=2015-11-03*
caq_async.ini	*>=2015-11-03*

Use the following settings to have the terminal input processed asynchronously by the server. Immediately after the processing request, the server sends an "OK" status and only then starts processing the data. Therefore, the user can go on and record further data, while the system is saving the previously collected data.

Error processing

If data is collected asynchronously, the user is not directly informed about errors. Only on the MOC, you can view error messages sent by the server: go to System administration --> Logging --> Dialog error logs.

1.16.1 Configuration

Configure the asynchronous processing in the separate INI file "caq_async.ini" in the directory \functions.

The following configuration settings are supported:

- Asynchronous processing of all dialogs and actions released by MPDV:

  To do so, set the option "ENABLE_ASYNC“ = "ON“ in the section "[SYSTEM]“.

- If the global option "ENABLE_ASYNC" is set, you can still disable the asynchronous processing for each dialog and action. To do so, create a section with the dialog name. Set the option to "= OFF" for all dialog actions you do not want to process asynchronously. The below example disables the action "CPAUMW.INSERT" in the dialog "QEE_MW_ME_ES_PP_SI“:

```
[System]
ENABLE_ASYNC=ON

[QEE_MW_ME_ES_PP_SI]
CPAUMW.INSERT=OFF
```

Note

MPDV has defined the dialogs and actions that are allowed to be processed asynchronously (see section Supported dialogs and actions).

1.16.2 Supported dialogs and actions

Currently, you can enable/disable the asynchronous processing for the following dialogs and actions:

Dialog	Action
QEE_MASS_CLASSIC	CMASSN.INSERT
QEE_ERR_CLASSIC	CPAUERR.INSERT
QEE_MM_BE_ST_PP_RA	CPAUERR.INSERT
QEE_MM_BE_ST_FS	CPAUERR.INSERT
QEE_MM_BE_ST_FS	CPAUERR.DELETE
QEE_MM_BE_ST_PP_FS	CPAUERR.INSERT

QEE_MM_BE_ST_PP_FS	CPAUERR.DELETE
QEE_MM_BE_ST_PP_SI	CPAUMW.INSERT
QEE_MM_BE_ST_SI	CPAUMW.INSERT
QEE_MM_BE_ST_SI	CPAUMW.UPDATE
QEE_MM_CO_ST_PP_SI	CPAUMW.INSERT
QEE_MW_ME_ES_PP_SI	CPAUMW.INSERT
QEE_MW_ME_ES_ST_SI	CPAUMW.INSERT
QEE_MW_ME_ES_ST_SI	CPAUMW.UPDATE
QEE_MM_BE_ST_PP_SI	CPAUMW.UPDATE
QEE_MM_CO_ST_PP_SI	CPAUMW.MODIFY
QEE_MW_ME_ES_PP_SI	CPAUMW.UPDATE
QEE_INSPPOINT	CPANUMP.ABSCHLIESSEN
QEE_INSPPOINT	CPANUMP.UPDATE
QEE_INSPPOINT_DETAIL	CPANUMP.ABSCHLIESSEN
QEE_INSPPOINT_DETAIL	CPANUMP.UPDATE

1.17 Calculated characteristic including calculation of eigenvalue (as of CAQ 8.2)

mpdv-aip.zip	*>=2015-11-05*
mpdv-aip.zip	*>=2015-11-05*
aip_qm.dlg	*>=2015-11-05*

This function is compatible with the option "save measured value with the ENTER key" and the "asynchronous processing" of inspection data.

You can use the following input types / workflows:

- Input type: `MESSW_ESTCK_PPUNKT_CALC (characteristic node)`

  Workflows: `WF: MM_ME_ES_PP_CA / DYN DLG AIP: QEE_MM_ME_ES_PP_CA`

- Input type `MESSW_ESTCK_STICHPR_CALC (characteristic node)`

  Workflows: `MM_ME_ES_SI_CA / DYN DLG AIP: QEE_MM_ME_ES_SI_CA`

- Input type `MESSW_ESTCK_PPUNKT_CALC (characteristic node)`

  Workflows: `MW_ME_ES_PP_CA / DYN DLG AIP: QEE_MW_ME_ES_PP_CA`

  Workflows: `MW_ME_ES_PP_CA / DYN DLG AIP: QEE_ERR_CLASSIC`

  Workflows: `MW_ME_ES_PP_CA / DYN DLG AIP: QEE_MASS_CLASSIC`

- Input type `MESSW_ESTCK_STICHPR_CALC (characteristic node)`

  Workflows: `MW_ME_ES_SI_CA / DYN DLG AIP: QEE_MW_ME_ES_SI_CA`

  Workflows: `MW_ME_ES_SI_CA / DYN DLG AIP: QEE_ERR_CLASSIC`

  Workflows: `MW_ME_ES_SI_CA / DYN DLG AIP: QEE_MASS_CLASSIC`

Use these input types to enable the recording of calculated characteristics including calculation of eigenvalue.

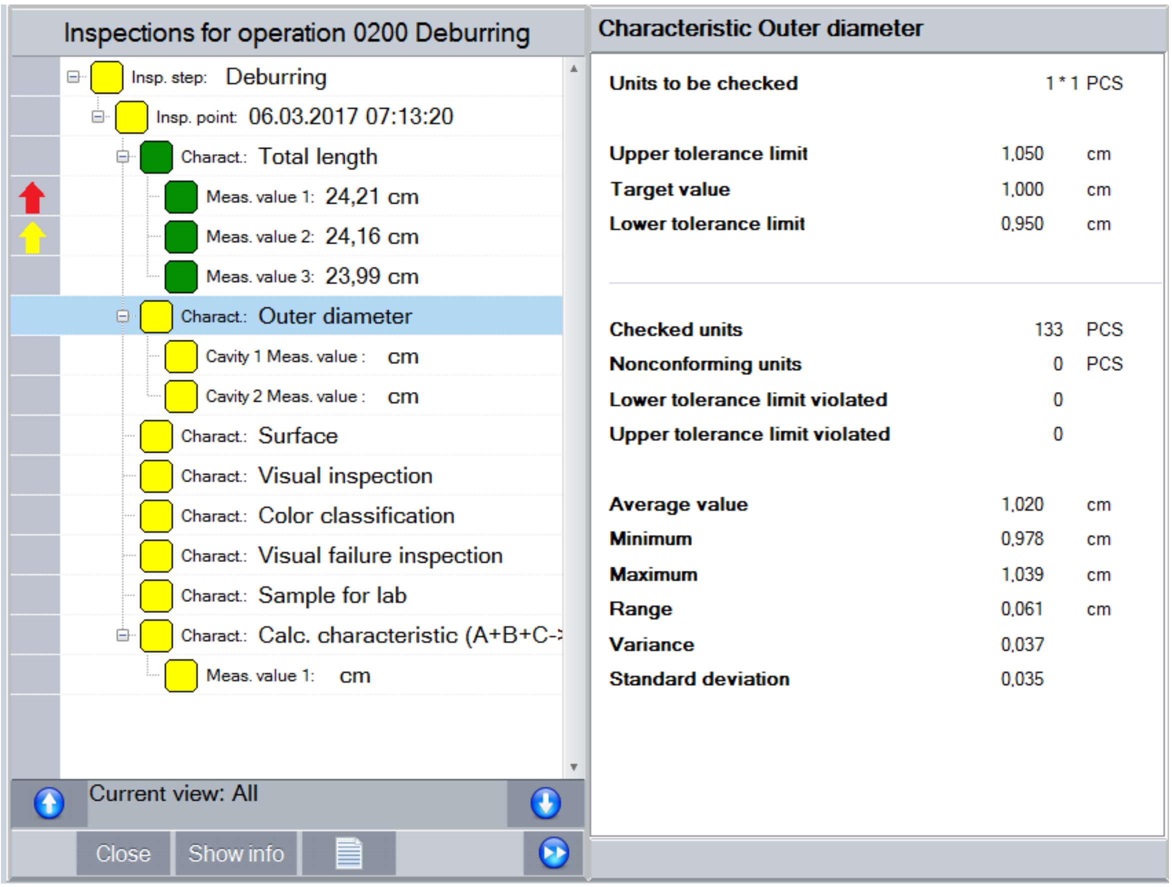


Figure: Input function "Calculated characteristics including calculation of eigenvalue" with reference to the inspection point.

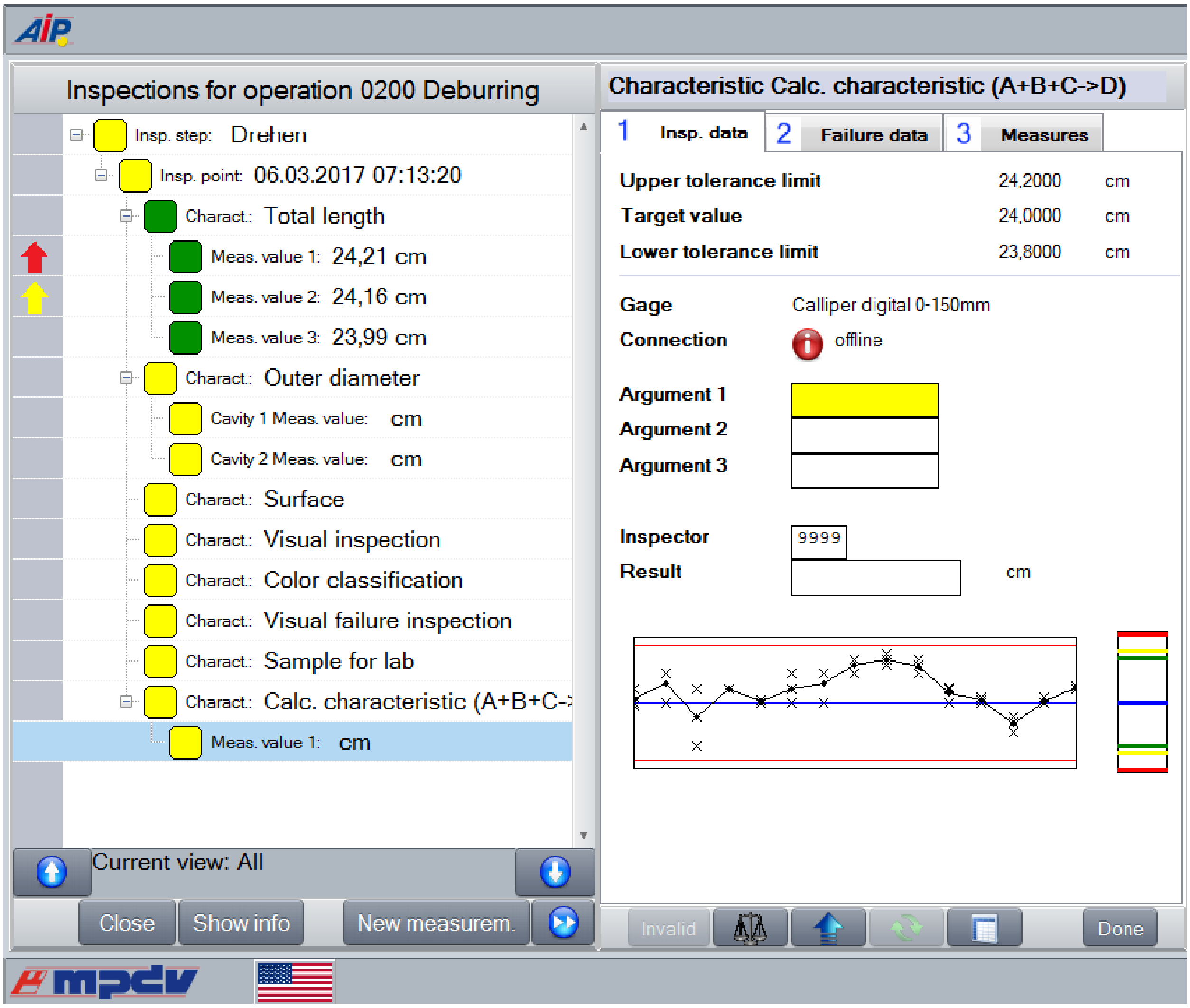


Figure: Input function "Calculated characteristics including calculation of eigenvalue" for a single value with reference to the inspection point.

The functions of the sample-related input type are identical to the functions of the input type relating to inspection points.

But here, the sample-related data collection is performed for the respective (machine-related) sample.

Functions / special features:

Field description

- The field *Gage* shows the test equipment used for the characteristic.

- The field *Connection* indicates the current (connection) status of the *test equipment.*

- The fields *Argument 1 to 4* are dynamically displayed or hidden. The fields are displayed, if they are included in the calculation formula of the characteristic. The fields are hidden if they are not included. By default, the field is empty. You can enter values that equal **0.**
  The field does not show the unit of the characteristic.
  The standard MDI function is available for these fields.

- The field *Result* shows the calculated value.

- The *Indicator of measured values* is linked to the field *Result* and always responds if the field *Result* includes a value. The field is read-only.

- Initially, the first visible "argument field" has the focus.

Start the calculation (button)

- If the dialog shows an argument field, then it is a mandatory field.
- Depending on the configuration, the field "inspector" is a mandatory field.
- You can calculate the result by clicking the *"calculate"* button. Then the field "result" shows the value. But the value is not yet saved.

Saving *inspection results:*

- If the dialog shows an argument field, then it is a mandatory field.
- Depending on the configuration, the field "inspector" is a mandatory field.
- You can save the collected data by clicking the button "*Done".*
- You can save the collected data by clicking the button "*Next".*

If required, you can change the dialog to include up to 10 *argument fields.* You can also add a *comment field* by changing the dialog configuration.

If the MDI connection is active, the system requests all MDI driver values of the corresponding channel without setting filter criteria.

In contrast to other characteristics, the system enters the MDI measured value directly into the field that has the focus. The focus switches to the next argument field, if the MDI measured value is confirmed `(CONFIRMED=1)`. If you are in the last argument field, the focus remains in this last field. The following options are available:

- Press the ENTER key
- Click the button "Done"
- Click the button 

1.18 Check if complaints exist when operations are logged on (as of SP13)

When you log on operations on the AIP, the system can check if complaints are available for the article to be produced that is specified in the order header. To activate this function, manually create the CAQ option 1219. The option 1219 defines the complaint types, complaint results and complaint statuses, which are checked. In the option, you also define the period of time that is checked. For details, refer to the procedure document of the option documentation "Configuration_QM_Options".

When creating the option 1219 with the value "Y", the system uses the configuration parameters in the "Addition" field during OP logon to check if complaints exist for the article or article + article index of the production order (order header). If complaints are found, then they are displayed in a message.